# 3.1 Limits and Continuity

**INTRODUCTION** The most important concept in elementary calculus is that of the limit. Recall that $\lim_{x \to x_0} f(x) = L$ intuitively means that values $f(x)$ of the function $f$ can be made arbitrarily close to the real number $L$ if values of $x$ are chosen sufficiently close to, but not equal to, the real number $x_0$. In real analysis, the concepts of continuity, the derivative, and the definite integral were *all* defined using the concept of a limit. Complex limits play an equally important role in study of complex analysis. The concept of a complex limit is similar to that of a real limit in the sense that $\lim_{z \to z_0} f(z) = L$ will mean that the values $f(z)$ of the complex function $f$ can be made arbitrarily close to the complex number $L$ if values of $z$ are chosen sufficiently close to, but not equal to, the complex number $z_0$. Although outwardly similar, there is an important difference between these two concepts of limit. In a real limit, there are two directions from which $x$ can approach $x_0$ on the real line, namely, from the left or from the right. In a complex limit, however, there are infinitely many directions from which $z$ can approach $z_0$ in the complex plane. In order for a complex limit to exist, each way in which $z$ can approach $z_0$ must yield the same limiting value.

In this section we will define the limit of a complex function, examine some of its properties, and introduce the concept of continuity for functions of a complex variable.

## 3.1.1 Limits

**Real Limits** The description of a real limit given in the section introduction is only an intuitive definition of this concept. In order to give the rigorous definition of a real limit, we must precisely state what is meant by the phrases "arbitrarily close to" and "sufficiently close to." The first thing to recognize is that a precise statement of these terms uses absolute values since $|a - b|$ measures the distance between two points on the real number line. On the real line, the points $x$ and $x_0$ are "close" if $|x - x_0|$ is a small positive number. Similarly, the points $f(x)$ and $L$ are "close" if $|f(x) - L|$ is a small positive number. In mathematics, it is customary to let the Greek letters $\varepsilon$ and $\delta$ represent small positive real numbers. Hence, the expression "$f(x)$ can be made arbitrarily close to $L$" can be made precise by stating that for any real number $\varepsilon > 0$, $x$ can be chosen so that $|f(x) - L| < \varepsilon$. In our intuitive definition we require that $|f(x) - L| < \varepsilon$ whenever values of $x$ are "sufficiently close to, but not equal to, $x_0$." This means that there is some distance $\delta > 0$ with the property that if $x$ is within distance $\delta$ of $x_0$ and $x \neq x_0$, then $|f(x) - L| < \varepsilon$. In other words, if $0 < |x - x_0| < \delta$, then $|f(x) - L| < \varepsilon$. The real number $\delta$ is not unique and, in general, depends on the choice of $\varepsilon$, the function $f$, and the point $x_0$. In summary, we have the following precise definition of the real limit:

A review of real limits can be found in Section 3.1 of the student study guide. ▶

*The limit of $f$ as $x$ tends to $x_0$ exists and is equal to $L$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - x_0| < \delta$.* (1)

The geometric interpretation of (1) is shown in FIGURE 3.1.1. In this figure we see that the graph of the function $y = f(x)$ over the interval $(x_0 - \delta, x_0 + \delta)$, excluding the point $x_0$, lies between the lines $y = L - \varepsilon$ and $y = L + \varepsilon$ shown dashed in Figure 3.1.1. In the terminology of mappings, the interval $(x_0 - \delta,\ x_0 + \delta)$, excluding the point $x = x_0$, shown in color on the $x$-axis is mapped onto the set shown in black in the interval $(L - \varepsilon,\ L + \varepsilon)$ on the $y$-axis. For the limit to exist, the relationship

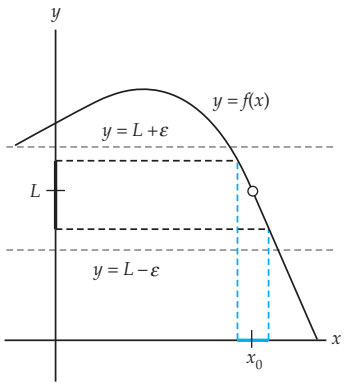
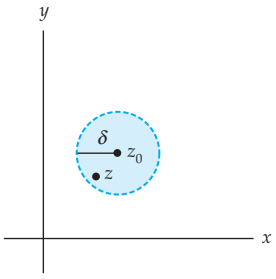
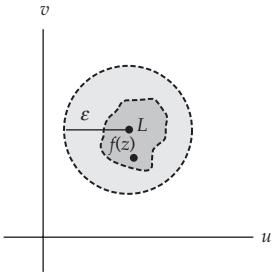



**FIGURE 3.1.1** Geometric meaning of a real limit




**(a)** Deleted $\delta$-neighborhood of $z_0$




**(b)** $\varepsilon$-neighborhood of $L$

**FIGURE 3.1.2** The geometric meaning of a complex limit

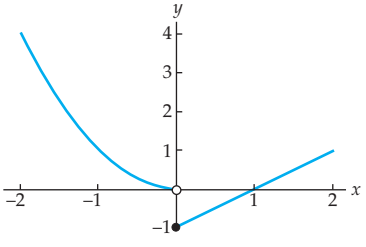



**FIGURE 3.1.3** The limit of $f$ does not exist as $x$ approaches 0

exhibited in Figure 3.1.1 must exist for any choice of $\varepsilon > 0$. We also see in Figure 3.1.1 that if a smaller $\varepsilon$ is chosen, then a smaller $\delta$ may be needed.

**Complex Limits** A complex limit is, in essence, the same as a real limit except that it is based on a notion of "close" in the complex plane. Because the distance in the complex plane between two points $z_1$ and $z_2$ is given by the modulus of the difference of $z_1$ and $z_2$, the precise definition of a complex limit will involve $|z_2 - z_1|$. For example, the phrase "$f(z)$ can be made arbitrarily close to the complex number $L$," can be stated precisely as: for every $\varepsilon > 0$, $z$ can be chosen so that $|f(z) - L| < \varepsilon$. Since the modulus of a complex number is a *real* number, both $\varepsilon$ and $\delta$ still represent small positive *real* numbers in the following definition of a complex limit. The complex analogue of (1) is:

**DEFINITION 3.1.1 Limit of a Complex Function**

Suppose that a complex function $f$ is defined in a deleted neighborhood of $z_0$ and suppose that $L$ is a complex number. The **limit of $f$ as $z$ tends to $z_0$ exists and is equal to $L$**, written as $\lim_{z \to z_0} f(z) = L$, if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$.

For complex functions, we rely on the concept of complex mappings to gain a geometric understanding of Definition 3.1.1. Recall from Section 1.5 that the set of points $w$ in the complex plane satisfying $|w - L| < \varepsilon$ is called a neighborhood of $L$, and that this set consists of all points in the complex plane lying within, but not on, a circle of radius $\varepsilon$ centered at the point $L$. Also recall that the set of points satisfying the inequalities $0 < |z - z_0| < \delta$ is called a deleted neighborhood of $z_0$ and consists of all points in the neighborhood $|z - z_0| < \delta$ excluding the point $z_0$. By Definition 3.1.1, if $\lim_{z \to z_0} f(z) = L$ and if $\varepsilon$ is any positive number, then there is a deleted neighborhood of $z_0$ of radius $\delta$ with the property that for every point $z$ in this deleted neighborhood, $f(z)$ is in the $\varepsilon$ neighborhood of $L$. That is, $f$ maps the deleted neighborhood $0 < |z - z_0| < \delta$ in the $z$-plane into the neighborhood $|w - L| < \varepsilon$ in the $w$-plane. In FIGURE 3.1.2(a), the deleted neighborhood of $z_0$ shown in color is mapped onto the set shown in dark gray in Figure 3.1.2(b). As required by Definition 3.1.1, the image lies within the $\varepsilon$-neighborhood of $L$ shown in light gray in Figure 3.1.2(b).

Complex and real limits have many common properties, but there is at least one very important difference. For real functions, $\lim_{x \to x_0} f(x) = L$ if and only if $\lim_{x \to x_0^+} f(x) = L$ and $\lim_{x \to x_0^-} f(x) = L$. That is, there are two directions from which $x$ can approach $x_0$ on the real line, from the right (denoted by $x \to x_0^+$) or from the left (denoted by $x \to x_0^-$). The real limit exists if and only if these two one-sided limits have the same value. For example, consider the real function defined by:

$$f(x) = \begin{cases} x^2, & x < 0 \\ x - 1, & x \geq 0 \end{cases}.$$

The limit of $f$ as $x$ approaches to 0 does not exist since $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} x^2 = 0$, but $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (x - 1) = -1$. See FIGURE 3.1.3.

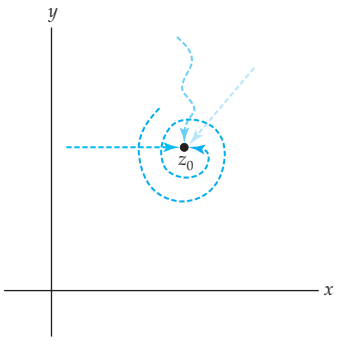



**FIGURE 3.1.4** Different ways to approach $z_0$ in a limit

For limits of complex functions, $z$ is allowed to approach $z_0$ from *any* direction in the complex plane, that is, along any curve or path through $z_0$. See **FIGURE 3.1.4**. In order that $\lim_{z\to z_0} f(z)$ exists and equals $L$, we require that $f(z)$ approach the same complex number $L$ along every possible curve through $z_0$. Put in a negative way:

### Criterion for the Nonexistence of a Limit

*If $f$ approaches two complex numbers $L_1 \neq L_2$ along two different curves or paths through $z_0$, then $\lim_{z\to z_0} f(z)$ does not exist.*

## EXAMPLE 1 A Limit That Does Not Exist

Show that $\lim_{z\to 0} \dfrac{z}{\bar{z}}$ does not exist.

**Solution** We show that this limit does not exist by finding two different ways of letting $z$ approach 0 that yield different values for $\lim_{z\to 0} \dfrac{z}{\bar{z}}$. First, we let $z$ approach 0 along the real axis. That is, we consider complex numbers of the form $z = x + 0i$ where the real number $x$ is approaching 0. For these points we have:

$$\lim_{z\to 0} \frac{z}{\bar{z}} = \lim_{x\to 0} \frac{x + 0i}{x - 0i} = \lim_{x\to 0} 1 = 1. \tag{2}$$

On the other hand, if we let $z$ approach 0 along the imaginary axis, then $z = 0 + iy$ where the real number $y$ is approaching 0. For this approach we have:

$$\lim_{z\to 0} \frac{z}{\bar{z}} = \lim_{y\to 0} \frac{0 + iy}{0 - iy} = \lim_{y\to 0} (-1) = -1. \tag{3}$$

Since the values in (2) and (3) are not the same, we conclude that $\lim_{z\to 0} \dfrac{z}{\bar{z}}$ does not exist. ☰

The limit $\lim_{z\to 0} \dfrac{z}{\bar{z}}$ from Example 1 did not exist because the values of $\lim_{z\to 0} \dfrac{z}{\bar{z}}$ as $z$ approached 0 along the real and imaginary axes did not agree. However, even if these two values did agree, the complex limit may still fail to exist. See Problems 19 and 20 in Exercises 3.1. In general, computing values of $\lim_{z\to z_0} f(z)$ as $z$ approaches $z_0$ from different directions, as in Example 1, can prove that a limit *does not exist*, but this technique cannot be used to prove that a limit *does exist*. In order to prove that a limit does exist we must use Definition 3.1.1 directly. This requires demonstrating that for every positive real number $\varepsilon$ there is an appropriate choice of $\delta$ that meets the requirements of Definition 3.1.1. Such proofs are commonly called "epsilon-delta proofs." Even for relatively simple functions, epsilon-delta proofs can be quite involved. Since this is an introductory text, we restrict our attention to what, in our opinion, are straightforward examples of epsilon-delta proofs.

## EXAMPLE 2 An Epsilon-Delta Proof of a Limit

Prove that $\lim_{z\to 1+i} (2 + i)z = 1 + 3i$.

**Solution** According to Definition 3.1.1, $\lim_{z\to 1+i} (2 + i)z = 1 + 3i$, if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|(2 + i)z - (1 + 3i)| < \varepsilon$ whenever $0 < |z - (1 + i)| < \delta$. Proving that the limit exists requires that we find an appropriate value of $\delta$ for a given

value of $\varepsilon$. In other words, for a given value of $\varepsilon$ we must find a positive number $\delta$ with the property that if $0 < |z-(1+i)| < \delta$, then $|(2+i)z-(1+3i)| < \varepsilon$. One way of finding $\delta$ is to "work backward." The idea is to start with the inequality:

$$|(2+i)z-(1+3i)| < \varepsilon \tag{4}$$

and then use properties of complex numbers and the modulus to manipulate this inequality until it involves the expression $|z-(1+i)|$. Thus, a natural first step is to factor $(2+i)$ out of the left-hand side of (4):

$$|2+i| \cdot \left| z - \frac{1+3i}{2+i} \right| < \varepsilon. \tag{5}$$

Because $|2+i| = \sqrt{5}$ and $\dfrac{1+3i}{2+i} = 1+i$, (5) is equivalent to:

$$\sqrt{5} \cdot |z-(1+i)| < \varepsilon \quad \text{or} \quad |z-(1+i)| < \frac{\varepsilon}{\sqrt{5}}. \tag{6}$$

Thus, (6) indicates that we should take $\delta = \varepsilon/\sqrt{5}$. Keep in mind that the choice of $\delta$ is not unique. Our choice of $\delta = \varepsilon/\sqrt{5}$ is a result of the particular algebraic manipulations that we employed to obtain (6). Having found $\delta$ we now present the formal proof that $\lim_{z\to 1+i}(2+i)z = 1+3i$ that does not indicate how the choice of $\delta$ was made:

Given $\varepsilon > 0$, let $\delta = \varepsilon/\sqrt{5}$. If $0 < |z-(1+i)| < \delta$, then we have $|z-(1+i)| < \varepsilon/\sqrt{5}$. Multiplying both sides of the last inequality by $|2+i| = \sqrt{5}$ we obtain:

$$|2+i| \cdot |z-(1+i)| < \sqrt{5} \cdot \frac{\varepsilon}{\sqrt{5}} \quad \text{or} \quad |(2+i)z-(1+3i)| < \varepsilon.$$

Therefore, $|(2+i)z-(1+3i)| < \varepsilon$ whenever $0 < |z-(1+i)| < \delta$. So, according to Definition 3.1.1, we have proven that $\lim_{z\to 1+i}(2+i)z = 1+3i$. ☰

**Real Multivariable Limits** The epsilon-delta proof from Example 2 illustrates the important fact that although the theory of complex limits is based on Definition 3.1.1, this definition does not provide a convenient method for *computing* limits. We now present a practical method for *computing* complex limits in Theorem 3.1.1. In addition to being a useful computational tool, this theorem also establishes an important connection between the *complex* limit of $f(z) = u(x, y) + iv(x, y)$ and the *real* limits of the real-valued functions of two real variables $u(x, y)$ and $v(x, y)$. Since every complex function is completely determined by the real functions $u$ and $v$, it should not be surprising that the limit of a complex function can be expressed in terms of the real limits of $u$ and $v$.

Before stating Theorem 3.1.1, we recall some of the important concepts regarding limits of real-valued functions of two real variables $F(x, y)$. The following definition of $\lim_{(x,y)\to(x_0,y_0)} F(x, y) = L$ is analogous to both (1) and Definition 3.1.1.

### Limit of the Real Function $F(x, y)$

*The limit of $F$ as $(x, y)$ tends to $(x_0, y_0)$ exists and is equal to the real number $L$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|F(x, y) - L| < \varepsilon$ whenever $0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$.* (7)

The expression $\sqrt{(x-x_0)^2 + (y-y_0)^2}$ in (7) represents the distance between the points $(x, y)$ and $(x_0, y_0)$ in the Cartesian plane. Using (7), it is relatively easy

to prove that:

$$\lim_{(x,y)\to(x_0,y_0)} 1 = 1, \quad \lim_{(x,y)\to(x_0,y_0)} x = x_0, \text{ and } \lim_{(x,y)\to(x_0,y_0)} y = y_0. \tag{8}$$

If $\lim_{(x,y)\to(x_0,y_0)} F(x, y) = L$ and $\lim_{(x,y)\to(x_0,y_0)} G(x, y) = M$, then (7) can also be used to show:

$$\lim_{(x,y)\to(x_0,y_0)} cF(x,\ y) = cL, \quad c \text{ a real constant}, \tag{9}$$

$$\lim_{(x,y)\to(x_0,y_0)} (F(x,\ y) \pm G(x,\ y)) = L \pm M, \tag{10}$$

$$\lim_{(x,y)\to(x_0,y_0)} F(x,\ y) \cdot G(x,\ y) = L \cdot M, \tag{11}$$

and

$$\lim_{(x,y)\to(x_0,y_0)} \frac{F(x,\ y)}{G(x,\ y)} = \frac{L}{M}, \ M \neq 0. \tag{12}$$

Limits involving polynomial expressions in $x$ and $y$ can be easily computed using the limits in (8) combined with properties (9)–(12). For example,

$$\begin{aligned}\lim_{(x,y)\to(1,2)} \left(3xy^2 - y\right) &= 3\left(\lim_{(x,y)\to(1,2)} x\right)\left(\lim_{(x,y)\to(1,2)} y\right)\left(\lim_{(x,y)\to(1,2)} y\right) - \lim_{(x,y)\to(1,2)} y \\ &= 3 \cdot 1 \cdot 2 \cdot 2 - 2 = 10.\end{aligned}$$

In general, if $p(x,\ y)$ is a two-variable polynomial function, then (8)–(12) can be used to show that

$$\lim_{(x,y)\to(x_0,y_0)} p(x,\ y) = p(x_0,\ y_0). \tag{13}$$

If $p(x,\ y)$ and $q(x,\ y)$ are two-variable polynomial functions and $q(x_0,\ y_0) \neq 0$, then (13) and (12) give:

$$\lim_{(x,y)\to(x_0,y_0)} \frac{p(x, y)}{q(x, y)} = \frac{p(x_0, y_0)}{q(x_0, y_0)}. \tag{14}$$

We now present Theorem 3.1.1, which relates real limits of $u(x,\ y)$ and $v(x,\ y)$ with the complex limit of $f(z) = u(x,\ y) + iv(x,\ y)$. An epsilon-delta proof of Theorem 3.1.1 can be found in Appendix I.

**THEOREM 3.1.1 Real and Imaginary Parts of a Limit**

Suppose that $f(z) = u(x,\ y) + iv(x,\ y)$, $z_0 = x_0 + iy_0$, and $L = u_0 + iv_0$. Then $\lim_{z\to z_0} f(z) = L$ if and only if

$$\lim_{(x,y)\to(x_0,y_0)} u(x,\ y) = u_0 \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0.$$

Theorem 3.1.1 has many uses. First and foremost, it allows us to compute many complex limits by simply computing a pair of real limits.

## EXAMPLE 3 Using Theorem 3.1.1 to Compute a Limit

Use Theorem 3.1.1 to compute $\lim_{z\to 1+i} \left(z^2 + i\right)$.

**Solution** Since $f(z) = z^2 + i = x^2 - y^2 + (2xy + 1)i$, we can apply Theorem 3.1.1 with $u(x,\ y) = x^2 - y^2$, $v(x,\ y) = 2xy + 1$, and $z_0 = 1 + i$. Identifying $x_0 = 1$ and

$y_0 = 1$, we find $u_0$ and $v_0$ by computing the two real limits:

$$u_0 = \lim_{(x,y)\to(1,1)} \left(x^2 - y^2\right) \quad \text{and} \quad v_0 = \lim_{(x,y)\to(1,1)} (2xy + 1).$$

Since both of these limits involve only multivariable polynomial functions, we can use (13) to obtain:

$$u_0 = \lim_{(x,y)\to(1,1)} \left(x^2 - y^2\right) = 1^2 - 1^2 = 0$$

and

$$v_0 = \lim_{(x,y)\to(1,1)} (2xy + 1) = 2 \cdot 1 \cdot 1 + 1 = 3,$$

and so $L = u_0 + iv_0 = 0 + i(3) = 3i$. Therefore, $\lim_{z\to 1+i} \left(z^2 + i\right) = 3i$. ☰

In addition to computing specific limits, Theorem 3.1.1 is also an important theoretical tool that allows us to derive many properties of complex limits from properties of real limits. The following theorem gives an example of this procedure.

**THEOREM 3.1.2 Properties of Complex Limits**

Suppose that $f$ and $g$ are complex functions. If $\lim_{z\to z_0} f(z) = L$ and $\lim_{z\to z_0} g(z) = M$, then

(*i*) $\lim_{z\to z_0} cf(z) = cL$, $c$ a complex constant,

(*ii*) $\lim_{z\to z_0} (f(z) \pm g(z)) = L \pm M$,

(*iii*) $\lim_{z\to z_0} f(z) \cdot g(z) = L \cdot M$, and

(*iv*) $\lim_{z\to z_0} \dfrac{f(z)}{g(z)} = \dfrac{L}{M}$, provided $M \neq 0$.

**Proof of (*i*)** Each part of Theorem 3.1.2 follows from Theorem 3.1.1 and the analogous property (9)–(12). We will prove part (*i*) and leave the remaining parts as exercises.

Let $f(z) = u(x,\ y) + iv(x,\ y)$, $z_0 = x_0 + iy_0$, $L = u_0 + iv_0$, and $c = a + ib$. Since $\lim_{z\to z_0} f(z) = L$, it follows from Theorem 3.1.1 that

$$\lim_{(x,y)\to(x_0,y_0)} u(x,\ y) = u_0 \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} v(x,\ y) = v_0.$$

By (9) and (10), we have

$$\lim_{(x,y)\to(x_0,y_0)} (au(x,\ y) - bv(x,\ y)) = au_0 - bv_0$$

and

$$\lim_{(x,y)\to(x_0,y_0)} (bu(x,\ y) + av(x,\ y)) = bu_0 + av_0.$$

However, $\text{Re}\,(cf(z)) = au(x,\ y) - bv(x,\ y)$ and $\text{Im}\,(cf(z)) = bu(x,\ y) + av(x,\ y)$. Therefore, by Theorem 3.1.1,

$$\lim_{z\to z_0} cf(z) = au_0 - bv_0 + i\,(bu_0 + av_0) = cL.$$ ☰

Of course the results in Theorems 3.1.2(*ii*) and 3.1.2(*iii*) hold for any finite sum of functions or finite product of functions, respectively. After establishing a couple of

basic complex limits, we can use Theorem 3.1.2 to compute a large number of limits in a very direct manner. The two basic limits that we need are those of the **complex constant function** $f(z) = c$, where $c$ is a complex constant, and the **complex identity function** $f(z) = z$. In Problem 45 in Exercises 3.1 you will be asked to show that:

$$\lim_{z \to z_0} c = c, \quad c \text{ a complex constant,} \tag{15}$$

and

$$\lim_{z \to z_0} z = z_0. \tag{16}$$

The following example illustrates how these basic limits can be combined with the Theorem 3.1.2 to compute limits of complex rational functions.

## EXAMPLE 4 Computing Limits with Theorem 3.1.2

Use Theorem 3.1.2 and the basic limits (15) and (16) to compute the limits

(**a**) $\displaystyle\lim_{z \to i} \frac{(3+i)z^4 - z^2 + 2z}{z+1}$

(**b**) $\displaystyle\lim_{z \to 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i}$

**Solution** (**a**) By Theorem 3.1.2(*iii*) and (16), we have:

$$\lim_{z \to i} z^2 = \lim_{z \to i} z \cdot z = \left(\lim_{z \to i} z\right) \cdot \left(\lim_{z \to i} z\right) = i \cdot i = -1.$$

Similarly, $\lim_{z \to i} z^4 = i^4 = 1$. Using these limits, Theorems 3.1.2(*i*), 3.1.2(*ii*), and the limit in (16), we obtain:

$$\begin{aligned} \lim_{z \to i} \left((3+i)z^4 - z^2 + 2z\right) &= (3+i) \lim_{z \to i} z^4 - \lim_{z \to i} z^2 + 2 \lim_{z \to i} z \\ &= (3+i)(1) - (-1) + 2(i) \\ &= 4 + 3i, \end{aligned}$$

and $\lim_{z \to i} (z+1) = 1 + i$. Therefore, by Theorem 3.1.2(*iv*), we have:

$$\lim_{z \to i} \frac{(3+i)z^4 - z^2 + 2z}{z+1} = \frac{\lim_{z \to i} \left((3+i)z^4 - z^2 + 2z\right)}{\lim_{z \to i} (z+1)} = \frac{4+3i}{1+i}.$$

After carrying out the division, we obtain $\displaystyle\lim_{z \to i} \frac{(3+i)z^4 - z^2 + 2z}{z+1} = \frac{7}{2} - \frac{1}{2}i$.

(**b**) In order to find $\displaystyle\lim_{z \to 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i}$, we proceed as in (a):

$$\begin{aligned} \lim_{z \to 1+\sqrt{3}i} \left(z^2 - 2z + 4\right) &= \left(1 + \sqrt{3}i\right)^2 - 2\left(1 + \sqrt{3}i\right) + 4 \\ &= -2 + 2\sqrt{3}i - 2 - 2\sqrt{3}i + 4 = 0, \end{aligned}$$

and $\lim_{z \to 1+\sqrt{3}i} \left(z - 1 - \sqrt{3}i\right) = 1 + \sqrt{3}i - 1 - \sqrt{3}i = 0$. It appears that we cannot apply Theorem 3.1.2(*iv*) since the limit of the denominator is 0. However, in the previous calculation we found that $1 + \sqrt{3}i$ is a root of the quadratic polynomial

$z^2 - 2z + 4$. From Section 1.6, recall that if $z_1$ is a root of a quadratic polynomial, then $z - z_1$ is a factor of the polynomial. Using long division, we find that

$$z^2 - 2z + 4 = (z - 1 + \sqrt{3}i)(z - 1 - \sqrt{3}i).$$

See (5) in Section 1.6. Because $z$ is not allowed to take on the value $1 + \sqrt{3}i$ in the limit, we can cancel the common factor in the numerator and denominator of the rational function. That is,

$$\lim_{z\to 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} = \lim_{z\to 1+\sqrt{3}i} \frac{(z - 1 + \sqrt{3}i)(z - 1 - \sqrt{3}i)}{z - 1 - \sqrt{3}i}$$

$$= \lim_{z\to 1+\sqrt{3}i} (z - 1 + \sqrt{3}i).$$

By Theorem 3.1.2(*ii*) and the limits in (15) and (16), we then have

$$\lim_{z\to 1+\sqrt{3}i} (z - 1 + \sqrt{3}i) = 1 + \sqrt{3}i - 1 + \sqrt{3}i = 2\sqrt{3}i.$$

Therefore, $\lim\limits_{z\to 1+\sqrt{3}i} \dfrac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} = 2\sqrt{3}i.$ ☰

In Section 3.2 we will calculate the limit in part (b) of Example 4 in a different manner.

### 3.1.2 Continuity

**Continuity of Real Functions** Recall that if the limit of a real function $f$ as $x$ approaches the point $x_0$ exists and agrees with the value of the function $f$ at $x_0$, then we say that $f$ is *continuous* at the point $x_0$. In symbols, this definition is given by:

**Continuity of a Real Function $f(x)$**

*A function $f$ is continuous at a point $x_0$ if* $\lim\limits_{x\to x_0} f(x) = f(x_0)$. (17)

Observe that in order for the equation $\lim\limits_{x\to x_0} f(x) = f(x_0)$ in (17) to be satisfied, three things must be true. The limit $\lim\limits_{x\to x_0} f(x)$ must exist, $f$ must be defined at $x_0$, and these two values must be equal. If any one of these three conditions fail, then $f$ cannot be continuous at $x_0$. For example, the function

$$f(x) = \begin{cases} x^2, & x < 0 \\ x - 1, & x \geq 0 \end{cases},$$

illustrated in Figure 3.1.3 is not continuous at the point $x = 0$ since $\lim\limits_{x\to 0} f(x)$ does not exist. On a similar note, even though $\lim\limits_{x\to 1} \dfrac{x^2 - 1}{x - 1} = 2$, the function $f(x) = \dfrac{x^2 - 1}{x - 1}$ is not continuous at $x = 1$ because $f(1)$ is not defined.

In real analysis, we visualize the concept of continuity using the graph of the function $f$. Informally, a function $f$ is continuous if there are no breaks or holes

in the graph of $f$. Since we cannot graph a complex function, our discussion of the continuity of complex functions will be primarily algebraic in nature.

**Continuity of Complex Functions** The definition of continuity for a complex function is, in essence, the same as that for a real function. That is, a complex function $f$ is continuous at a point $z_0$ if the limit of $f$ as $z$ approaches $z_0$ exists and is the same as the value of $f$ at $z_0$. This gives the following definition for complex functions, which is analogous to (17).

**DEFINITION 3.1.2 Continuity of a Complex Function**

A complex function $f$ is **continuous at a point** $z_0$ if

$$\lim_{z \to z_0} f(z) = f(z_0).$$

The definition of continuity can also be given using epsilons and deltas:

*A function $f$ is continuous at $z_0$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - f(z_0)| < \varepsilon$ whenever $0 < |z - z_0| < \delta$.*

This epsilon-delta definition will be used to establish certain theorems on integration in Chapter 5. Analogous to real functions, if a complex $f$ is continuous at a point, then the following three conditions must be met.

**Criteria for Continuity at a Point**

*A complex function $f$ is continuous at a point $z_0$ if each of the following three conditions hold:*

(*i*) $\lim_{z \to z_0} f(z)$ *exists,*

(*ii*) $f$ *is defined at* $z_0$, *and*

(*iii*) $\lim_{z \to z_0} f(z) = f(z_0)$.

If a complex function $f$ is not continuous at a point $z_0$ then we say that $f$ is **discontinuous** at $z_0$. For example, the function $f(z) = \dfrac{1}{1+z^2}$ is discontinuous at $z = i$ and $z = -i$.

**EXAMPLE 5 Checking Continuity at a Point**

Consider the function $f(z) = z^2 - iz + 2$. In order to determine if $f$ is continuous at, say, the point $z_0 = 1 - i$, we must find $\lim_{z \to z_0} f(z)$ and $f(z_0)$, then check to see whether these two complex values are equal. From Theorem 3.1.2 and the limits in (15) and (16) we obtain:

$$\lim_{z \to z_0} f(z) = \lim_{z \to 1-i} \left(z^2 - iz + 2\right) = (1-i)^2 - i(1-i) + 2 = 1 - 3i.$$

Furthermore, for $z_0 = 1 - i$ we have:

$$f(z_0) = f(1-i) = (1-i)^2 - i(1-i) + 2 = 1 - 3i.$$

Since $\lim_{z \to z_0} f(z) = f(z_0)$, we conclude that $f(z) = z^2 - iz + 2$ is continuous at the point $z_0 = 1 - i$. ☰

As Example 5 indicates, the continuity of a complex polynomial or a rational function is easily determined using Theorem 3.1.2 and the limits in (15) and (16). More complicated functions, however, often require other techniques.

### EXAMPLE 6 Discontinuity of Principal Square Root Function

Show that the principal square root function $f(z) = z^{1/2}$ defined by (8) of Section 2.4 is discontinuous at the point $z_0 = -1$.

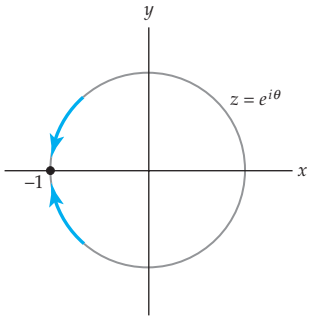



FIGURE 3.1.5 Figure for Example 6

**Solution** We show that $f(z) = z^{1/2} = \sqrt{r}e^{i\theta/2}$, $\theta = \text{Arg}(z)$, is discontinuous at $z_0 = -1$ by demonstrating that the limit as $z$ approaches $-1$ does not exist. We do so by finding two ways of letting $z$ approach $-1$ that yield different values of the limit. First, consider $z$ approaching $-1$ along the quarter of the unit circle lying in the second quadrant. See FIGURE 3.1.5. For this approach we have $z = e^{i\theta}$, $\pi/2 < \theta < \pi$, and $\theta$ approaching $\pi$. Thus, by setting $r = 1$ and letting $\theta$ approach $\pi$, we obtain:

$$\lim_{z\to -1} z^{1/2} = \lim_{z\to -1} \sqrt{r}e^{i\theta/2} = \lim_{\theta\to\pi} e^{i\theta/2}.$$

We evaluate this limit using Theorem 3.1.1:

$$\lim_{z\to -1} z^{1/2} = \lim_{\theta\to\pi} e^{i\theta/2} = \lim_{\theta\to\pi}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right) = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i. \tag{18}$$

Next, we let $z$ approach $-1$ along the quarter of the unit circle lying in the third quadrant. Again refer to Figure 3.1.5. Along this curve we have $z = e^{i\theta}$, $-\pi < \theta < -\pi/2$, and $\theta$ approaching $-\pi$. For these points:

$$\lim_{z\to -1} z^{1/2} = \lim_{\theta\to -\pi} e^{i\theta/2} = \lim_{\theta\to -\pi}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right) = -i. \tag{19}$$

Because the complex values in (18) and (19) do not agree, we conclude that $\lim_{z\to -1} z^{1/2}$ does not exist. Therefore, $f(z) = z^{1/2}$ is discontinuous at $z_0 = -1$. ☰

In Definition 3.1.2 we defined continuity of a complex function $f$ at a single point $z_0$ in the complex plane. We are often also interested in the continuity of a function on a set of points in the complex plane. A complex function $f$ is **continuous on a set $S$** if $f$ is continuous at $z_0$ for each $z_0$ in $S$. For example, using Theorem 3.1.2 and the limits in (15) and (16) we can show that $f(z) = z^2 - iz + 2$ is continuous at any point $z_0$ in the complex plane. Therefore, $f(z) = z^2 - iz + 2$ is continuous on $\mathbf{C}$. The function $f(z) = \dfrac{1}{z^2+1}$, on the other hand, is continuous on the set consisting of all complex $z$ such that $z \neq \pm i$.

### Properties of Continuous Functions

Because the concept of continuity is defined using the complex limit, various properties of complex limits can be translated into statements about continuity. The following result is an example of this.

**THEOREM 3.1.3 Real and Imaginary Parts of a Continuous Function**

If $f(z) = u(x, y) + iv(x, y)$ and $z_0 = x_0 + iy_0$, then the complex function $f$ is continuous at the point $z_0$ if and only if both real functions $u$ and $v$ are continuous at the point $(x_0, y_0)$.

**Proof** If $f$ is continuous at $z_0$, then from Definition 3.1.2 we have:

$$\lim_{z \to z_0} f(z) = f(z_0) = u(x_0, y_0) + iv(x_0, y_0).$$

After making the identifications $u_0 = u(x_0, y_0)$ and $v_0 = v(x_0, y_0)$, Theorem 3.1.1 gives:

$$\lim_{(x,y)\to(x_0,y_0)} u(x, y) = u(x_0, y_0) \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} v(x, y) = v(x_0, y_0).$$

Therefore, by the definition of continuity for real multivariable functions, both $u$ and $v$ are continuous at $(x_0, y_0)$. Conversely, if $u$ and $v$ are continuous at $(x_0, y_0)$, then

$$\lim_{(x,y)\to(x_0,y_0)} u(x, y) = u(x_0, y_0) \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} v(x, y) = v(x_0, y_0).$$

It then follows from Theorem 3.1.1 that $\lim_{z \to z_0} f(z) = u(x_0, y_0) + iv(x_0, y_0) = f(z_0)$. Therefore, $f$ is continuous by Definition 3.1.2. ☰

## EXAMPLE 7 Checking Continuity Using Theorem 3.1.3

Show that the function $f(z) = \bar{z}$ is continuous on $\mathbf{C}$.

**Solution** According to Theorem 3.1.3, $f(z) = \bar{z} = \overline{x + iy} = x - iy$ is continuous at $z_0 = x_0 + iy_0$ if both $u(x, y) = x$ and $v(x, y) = -y$ are continuous at $(x_0, y_0)$. Because $u$ and $v$ are two-variable polynomial functions, it follows from (13) that:

$$\lim_{(x,y)\to(x_0,y_0)} u(x, y) = x_0 \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} v(x, y) = -y_0.$$

This implies that $u$ and $v$ are continuous at $(x_0, y_0)$, and, therefore, that $f$ is continuous at $z_0 = x_0 + iy_0$. Since $z_0 = x_0 + iy_0$ was an arbitrary point, we conclude that the function $f(z) = \bar{z}$ is continuous on $\mathbf{C}$. ☰

The algebraic properties of complex limits from Theorem 3.1.2 can also be restated in terms of continuity of complex functions.

### THEOREM 3.1.4 Properties of Continuous Functions

If $f$ and $g$ are continuous at the point $z_0$, then the following functions are continuous at the point $z_0$:

(*i*) $cf$, $c$ a complex constant,

(*ii*) $f \pm g$,

(*iii*) $f \cdot g$, and

(*iv*) $\dfrac{f}{g}$, provided $g(z_0) \neq 0$.

**Proof of (*ii*)** We prove only (*ii*); proofs of the remaining parts are similar. Since $f$ and $g$ are continuous at $z_0$ we have that $\lim_{z \to z_0} f(z) = f(z_0)$ and $\lim_{z \to z_0} g(z) = g(z_0)$. From Theorem 3.1.2(*ii*), it follows that $\lim_{z \to z_0} (f(z) + g(z)) = f(z_0) + g(z_0)$. Therefore, $f + g$ is continuous at $z_0$ by Definition 3.1.2. ☰

Of course, the results of Theorems 3.1.4(*ii*) and 3.1.4(*iii*) extend to any finite sum or finite product of continuous functions, respectively. We can use these facts to show that complex polynomials are continuous functions.

**THEOREM 3.1.5 Continuity of Polynomial Functions**

Complex polynomial functions are continuous on the entire complex plane.

**Proof** Let $p(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$ be a complex polynomial function and let $z_0$ be any point in the complex plane $\mathbf{C}$. From (16) we have that the identity function $f(z) = z$ is continuous at $z_0$, and by repeated application of Theorem 3.1.4(*iii*), this implies that the power function $f(z) = z^n$, where $n$ is an integer and $n \geq 1$, is continuous at this point as well. Moreover, (15) implies that every complex constant function $f(z) = c$ is continuous at $z_0$, and so it follows from Theorem 3.1.4(*i*) that each of the functions $a_n z^n$, $a_{n-1}z^{n-1}$, $\ldots$, $a_1 z$, and $a_0$ are continuous at $z_0$. Now from repeated application of Theorem 3.1.4(*ii*) we see that $p(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$ is continuous at $z_0$. Since $z_0$ was allowed to be any point in the complex plane, we have shown that the polynomial function $p$ is continuous on the entire complex plane $\mathbf{C}$. ☰

Since a rational function $f(z) = p(z)/q(z)$ is quotient of the polynomial functions $p$ and $q$, it follows from Theorem 3.1.5 and Theorem 3.1.4(*iv*) that $f$ is continuous at every point $z_0$ for which $q(z_0) \neq 0$. In other words,

**Continuity of Rational Functions**

*Rational functions are continuous on their domains.*

**Bounded Functions** Continuous complex functions have many important properties that are analogous to properties of continuous real functions. For instance, if a real function $f$ is continuous on a closed interval $I$ on the real line, then $f$ is bounded on $I$. This means that there is a real number $M > 0$ such that $|f(x)| \leq M$ for all $x$ in $I$. A similar result for real multivariable functions states that if $F(x, y)$ is continuous on a closed and bounded region $R$ of the Cartesian plane, then there is a real number $M > 0$ such that $|F(x, y)| \leq M$ for all $(x, y)$ in $R$, and we say $F$ is bounded on $R$.

Now suppose that the complex function $f(z) = u(x, y) + iv(x, y)$ is defined on a closed and bounded region $R$ in the complex plane. As with real functions, we say that $f$ is **bounded** on $R$ if there exists a real constant $M > 0$ such that $|f(z)| < M$ for all $z$ in $R$. If $f$ is continuous on $R$, then Theorem 3.1.3 tells us that $u$ and $v$ are continuous real functions on $R$. It follows that $F(x, y) = \sqrt{[u(x, y)]^2 + [v(x, y)]^2}$ is also continuous on $R$. Because $F$ is continuous on the closed and bounded region $R$, we conclude that $F$ is bounded on $R$. That is, there is a real constant $M > 0$ such that $|F(x, y)| \leq M$ for all $(x, y)$ in $R$. However, since $|f(z)| = F(x, y)$, we have that $|f(z)| \leq M$ for all $z$ in $R$. In other words, the complex function $f$ is bounded on $R$. This establishes the following important property of continuous complex functions.

**A Bounding Property**

*If a complex function $f$ is continuous on a closed and bounded region $R$, then $f$ is bounded on $R$. That is, there is a real constant $M > 0$ such that $|f(z)| \leq M$ for all $z$ in $R$.*

While this result assures us that a bound $M$ exists for $f$ on $R$, it offers no practical approach to find it. One approach to find a bound is to use the triangle inequality. See

Example 3 in Section 1.2. Another approach to determine a bound is to use complex mappings. See Problems 38 and 39 in Exercises 2.3, Problems 55 and 56 in Exercises 2.4, and Problems 29 and 30 in Exercises 2.5. In Chapter 5, we will see that for a special class of important complex functions, the bound can only be attained by a point in the boundary of $R$.

**Branches** In Section 2.4 we discussed, briefly, the concept of a multiple-valued function $F(z)$ that assigns a set of complex numbers to the input $z$. (Recall that our convention is to always use uppercase letters such as $F$, $G$, and $H$ to represent multiple-valued functions.) Examples of multiple-valued functions include $F(z) = z^{1/n}$, which assigns to the input $z$ the set of $n$ $n$th roots of $z$, and $G(z) = \arg(z)$, which assigns to the input $z$ the infinite set of arguments of $z$. In practice, it is often the case that we need a consistent way of choosing just one of the roots of a complex number or, maybe, just one of the arguments of a complex number. That is, we are usually interested in computing just one of the values of a multiple-valued function. If we make this choice of value with the concept of continuity in mind, then we obtain a *function* that is called a *branch* of a multiple-valued function. In more rigorous terms, a **branch** of a multiple-valued function $F$ is a function $f_1$ that is continuous on some domain and that assigns exactly one of the multiple-values of $F$ to each point $z$ in that domain.

### Notation: Branches

*When representing branches of a multiple-valued function $F$ with functional notation, we will use lowercase letters with a numerical subscript such as $f_1$, $f_2$, and so on.*

The requirement that a branch be continuous means that the domain of a branch is different from the domain of the multiple-valued function. For example, the multiple-valued function $F(z) = z^{1/2}$ that assigns to each input $z$ the set of two square roots of $z$ is defined for all nonzero complex numbers $z$. Even though the principal square root function $f(z) = z^{1/2}$ does assign exactly one value of $F$ to each input $z$ (namely, it assigns to $z$ the principal square root of $z$), $f$ *is not* a branch of $F$. The reason for this is that the principal square root function is not continuous on its domain. In particular, in Example 6 we showed that $f(z) = z^{1/2}$ is not continuous at $z_0 = -1$. The argument used in Example 6 can be easily modified to show that $f(z) = z^{1/2}$ is discontinuous at every point on the negative real axis. Therefore, in order to obtain a branch of $F(z) = z^{1/2}$ that agrees with the principal square root function, we must restrict the domain to exclude points on the negative real axis. This gives the function

$$f_1(z) = \sqrt{r}e^{i\theta/2}, \quad -\pi < \theta < \pi. \tag{20}$$

We call the function $f_1$ defined by (20) the **principal branch** of $F(z) = z^{1/2}$ because the value of $\theta$ represents the principal argument of $z$ for all $z$ in $\text{Dom}(f_1)$. In the following example we show that $f_1$ is, in fact, a branch of $F$.

### EXAMPLE 8 A Branch of $F(z) = z^{1/2}$

Show that the function $f_1$ defined by (20) is a branch of the multiple-valued function $F(z) = z^{1/2}$.

**Solution** The domain of the function $f_1$ is the set defined by $|z| > 0$, $-\pi < \arg(z) < \pi$, shown in gray in FIGURE 3.1.6. From (8) of Section 2.4, we see

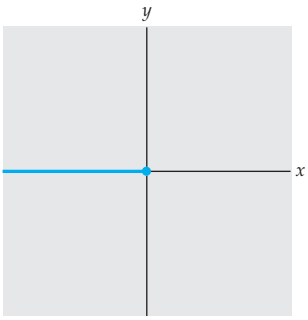
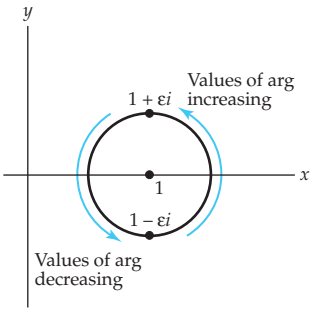



**FIGURE 3.1.6** The domain $D$ of the branch $f_1$

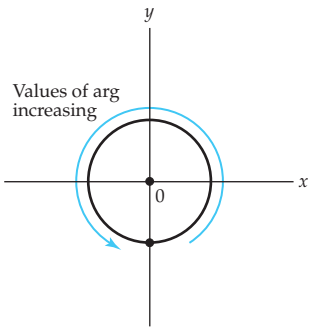
that the function $f_1$ agrees with the principal square root function $f$ on this set. Thus, $f_1$ does assign to the input $z$ exactly one of the values of $F(z) = z^{1/2}$. It remains to show that $f_1$ is a continuous function on its domain. In order to see that this is so, let $z$ be a point with $|z| > 0$, $-\pi < \arg(z) < \pi$. If $z = x + iy$ and $x > 0$, then $z = re^{i\theta}$ where $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1}(y/x)$. Since $-\pi/2 < \tan^{-1}(y/x) < \pi/2$, the inequality $-\pi < \theta < \pi$ is satisfied. Thus, substituting the expressions for $r$ and $\theta$ in (20) we obtain:

$$f_1(z) = \sqrt[4]{x^2 + y^2}\, e^{i \tan^{-1}(y/x)/2}$$
$$= \sqrt[4]{x^2 + y^2} \cos\left(\frac{\tan^{-1}(y/x)}{2}\right) + i\sqrt[4]{x^2 + y^2} \sin\left(\frac{\tan^{-1}(y/x)}{2}\right).$$

Because the real and imaginary parts of $f_1$ are continuous real functions for $x > 0$, we conclude from Theorem 3.1.3 that $f_1$ is continuous for $x > 0$. A similar argument can be made for points with $y > 0$ using $\theta = \cot^{-1}(x/y)$ and for points with $y < 0$ using $\theta = -\cot^{-1}(x/y)$. In each case, we conclude from Theorem 3.1.3 that $f_1$ is continuous. Therefore, the function $f_1$ defined in (20) is a branch of the multiple-valued function $F(z) = z^{1/2}$. ☰




**(a)** $z = 1$ is not a branch point




**(b)** $z = 0$ is a branch point

**FIGURE 3.1.7** $G(z) = \arg(z)$

**Branch Cuts and Points** Although the multiple-valued function $F(z) = z^{1/2}$ is defined for all nonzero complex numbers $\mathbf{C}$, the principal branch $f_1$ is defined only on the domain $|z| > 0$, $-\pi < \arg(z) < \pi$. In general, a **branch cut** for a branch $f_1$ of a multiple-valued function $F$ is a curve that is excluded from the domain of $F$ so that $f_1$ is continuous on the remaining points. The nonpositive real axis, shown in color in Figure 3.1.6, is a branch cut for the principal branch $f_1$ given by (20) of the multiple-valued function $F(z) = z^{1/2}$. A different branch of $F$ with the same branch cut is given by $f_2(z) = \sqrt{r}e^{i\theta/2}$, $\pi < \theta < 3\pi$. These branches are distinct because for, say, $z = i$ we have $f_1(i) = \frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}i$, but $f_2(i) = -\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}i$. Notice that if we set $\phi = \theta - 2\pi$, then the branch $f_2$ can be expressed as $f_2(z) = \sqrt{r}e^{i(\phi+2\pi)/2} = \sqrt{r}e^{i\phi/2}e^{i\pi}$, $-\pi < \phi < \pi$. Since $e^{i\pi} = -1$, this simplifies to $f_2(z) = -\sqrt{r}e^{i\phi/2}$, $-\pi < \phi < \pi$. Thus, we have shown that $f_2 = -f_1$. You can think of these two branches of $F(z) = z^{1/2}$ as being analogous to the positive and negative square roots of a positive real number. Other branches of $F(z) = z^{1/2}$ can be defined in a similar manner using any ray emanating from the origin as a branch cut. For example, $f_3(z) = \sqrt{r}e^{i\theta/2}$, $-3\pi/4 < \theta < 5\pi/4$, defines a branch of $F(z) = z^{1/2}$. The branch cut for $f_3$ is the ray $\arg(z) = -3\pi/4$ together with the point $z = 0$.

It is not a coincidence that the point $z = 0$ is on the branch cut for $f_1$, $f_2$, and $f_3$. The point $z = 0$ must be on the branch cut of *every* branch of the multiple-valued function $F(z) = z^{1/2}$. In general, a point with the property that it is on the branch cut of every branch is called a **branch point** of $F$. Alternatively, a branch point is a point $z_0$ with the following property: If we traverse any circle centered at $z_0$ with sufficiently small radius starting at a point $z_1$, then the values of any branch do not return to the value at $z_1$. For example, consider any branch of the multiple-valued function $G(z) = \arg(z)$. At the point, say, $z_0 = 1$, if we traverse the small circle $|z - 1| = \varepsilon$ counterclockwise from the point $z_1 = 1 - \varepsilon i$, then the values of the branch increase until we reach the point $1 + \varepsilon i$ and then the values of the branch decrease back down to the value of the branch at $z_1$. See FIGURE 3.1.7(a). This means that the point $z_0 = 1$ is not a branch point. On the other hand, suppose we repeat this process for the point $z_0 = 0$. For the small circle $|z| = \varepsilon$, the values of the branch increase along the entire circle. See Figure 3.1.7(b). By the time we have returned to our starting point, the value of the branch is no

longer the same; it has increased by $2\pi$. Therefore, $z_0 = 0$ is a branch point of $G(z) = \arg(z)$.

### *Remarks* COMPARISON WITH REAL ANALYSIS

(*i*) As with real analysis, we can define the concepts of infinite limits and limits at infinity for complex functions. Intuitively, the limit $\lim_{z\to\infty} f(z) = L$ means that values of the function $f$ can be made arbitrarily close to $L$ if values of $z$ are chosen so that $|z|$ is sufficiently large. A precise statement of a **limit at infinity** is:

*The limit of $f$ as $z$ tends to $\infty$ exists and is equal to $L$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $|z| > 1/\delta$.*

Using this definition it is not hard to show that:

$$\lim_{z\to\infty} f(z) = L \text{ if and only if } \lim_{z\to 0} f\left(\frac{1}{z}\right) = L. \tag{21}$$

Similarly, the **infinite limit** $\lim_{z\to z_0} f(z) = \infty$ is defined by:

*The limit of $f$ as $z$ tends to $z_0$ is $\infty$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(z)| > 1/\varepsilon$ whenever $0 < |z - z_0| < \delta$.*

From this definition we obtain the following result:

$$\lim_{z\to z_0} f(z) = \infty \text{ if and only if } \lim_{z\to z_0} \frac{1}{f(z)} = 0. \tag{22}$$

See Problems 21–26 in Exercises 3.1.

(*ii*) In real analysis we visualize a continuous function as a function whose graph has no breaks or holes in it. It is natural to ask if there is an analogous property for continuous complex functions. The answer is yes, but this property must be stated in terms of complex mappings. We begin by recalling that a parametric curve defined by parametric equations $x = x(t)$ and $y = y(t)$ is called continuous if the real functions $x$ and $y$ are continuous. In a similar manner, we say that a complex parametric curve defined by $z(t) = x(t) + iy(t)$ is **continuous** if both $x(t)$ and $y(t)$ are continuous real functions. As with parametric curves in the Cartesian plane, a continuous parametric curve in the complex plane has no breaks or holes in it. Such curves provide a means to visualize continuous complex functions.

*If a complex function $f$ is continuous on a set $S$, then the image of every continuous parametric curve in $S$ must be a continuous curve.*

To see why this is so, consider a continuous complex function $f(z) = u(x,\ y) + iv(x,\ y)$ and a continuous parametric curve defined by $z(t) = x(t) + iy(t)$. From Theorem 3.1.3, $u(x,\ y)$ and $v(x,\ y)$ are continuous real functions. Moreover, since $x(t)$ and $y(t)$ are continuous functions, it follows from multivariable calculus that the compositions $u(x(t),\ y(t))$ and $v(x(t),\ y(t))$ are continuous functions. Therefore, the image of the parametric curve given by $w(t) = f(z(t)) = u(x(t),\ y(t)) + iv(x(t),\ y(t))$ is continuous. See Problems 59–62 in Exercises 3.1.

## EXERCISES 3.1

Answers to selected odd-numbered problems begin on page ANS-10.

### 3.1.1 Limits

In Problems 1–8, use Theorem 3.1.1 and the properties of real limits on page 103 to compute the given complex limit.

**1.** $\lim_{z\to 2i}\left(z^2-\bar{z}\right)$

**2.** $\lim_{z\to 1+i}\dfrac{z-\bar{z}}{z+\bar{z}}$

**3.** $\lim_{z\to 1-i}\left(|z|^2-i\bar{z}\right)$

**4.** $\lim_{z\to 3i}\dfrac{\text{Im}\left(z^2\right)}{z+\text{Re}\,(z)}$

**5.** $\lim_{z\to \pi i} e^z$

**6.** $\lim_{z\to 0}\dfrac{z^2+\bar{z}^2}{\text{Re}(z)+\text{Im}(z)}$

**7.** $\lim_{z\to 0}\dfrac{e^z-e^{\bar{z}}}{\text{Im}(z)}$

**8.** $\lim_{z\to 1+i}\left(\log_e\left|x^2+y^2\right|+i\arctan\dfrac{y}{x}\right)$

In Problems 9–16, use Theorem 3.1.2 and the basic limits (15) and (16) to compute the given complex limit.

**9.** $\lim_{z\to 2-i}\left(z^2-z\right)$

**10.** $\lim_{z\to i}\left(z^5-z^2+z\right)$

**11.** $\lim_{z\to e^{i\pi/4}}\left(z+\dfrac{1}{z}\right)$

**12.** $\lim_{z\to 1+i}\dfrac{z^2+1}{z^2-1}$

**13.** $\lim_{z\to -i}\dfrac{z^4-1}{z+i}$

**14.** $\lim_{z\to 2+i}\dfrac{z^2-(2+i)^2}{z-(2+i)}$

**15.** $\lim_{z\to z_0}\dfrac{(az+b)-(az_0+b)}{z-z_0}$

**16.** $\lim_{z\to -3+i\sqrt{2}}\dfrac{z+3-i\sqrt{2}}{z^2+6z+11}$

**17.** Consider the limit $\lim_{z\to 0}\dfrac{\text{Re}(z)}{\text{Im}(z)}$.

(**a**) What value does the limit approach as $z$ approaches 0 along the line $y=x$?

(**b**) What value does the limit approach as $z$ approaches 0 along the imaginary axis?

(**c**) Based on your answers for (a) and (b), what can you say about $\lim_{z\to 0}\dfrac{\text{Re}(z)}{\text{Im}(z)}$?

**18.** Consider the limit $\lim_{z\to i}(|z|+i\,\text{Arg}\,(iz))$.

(**a**) What value does the limit approach as $z$ approaches $i$ along the unit circle $|z|=1$ in the first quadrant?

(**b**) What value does the limit approach as $z$ approaches $i$ along the unit circle $|z|=1$ in the second quadrant?

(**c**) Based on your answers for (a) and (b), what can you say about $\lim_{z\to i}(|z|+i\,\text{Arg}\,(iz))$?

**19.** Consider the limit $\lim_{z\to 0}\left(\dfrac{z}{\bar{z}}\right)^2$.

(**a**) What value does the limit approach as $z$ approaches 0 along the real axis?

(**b**) What value does the limit approach as $z$ approaches 0 along the imaginary axis?

(**c**) Do the answers from (a) and (b) imply that $\lim_{z\to 0}\left(\dfrac{z}{\bar{z}}\right)^2$ exists? Explain.

(**d**) What value does the limit approach as $z$ approaches 0 along the line $y=x$?

(**e**) What can you say about $\lim_{z\to 0}\left(\dfrac{z}{\bar{z}}\right)^2$?

**20.** Consider the limit $\lim\limits_{z\to 0}\left(\dfrac{2y^2}{x^2}-\dfrac{x^2-y^2}{y^2}i\right)$.

(**a**) What value does the limit approach as $z$ approaches 0 along the line $y=x$?

(**b**) What value does the limit approach as $z$ approaches 0 along the line $y=-x$?

(**c**) Do the answers from (a) and (b) imply that $\lim\limits_{z\to 0}\left(\dfrac{2y^2}{x^2}-\dfrac{x^2-y^2}{y^2}i\right)$ exists? Explain.

(**d**) What value does the limit approach as $z$ approaches 0 along the line $y=2x$?

(**e**) What can you say about $\lim\limits_{z\to 0}\left(\dfrac{2y^2}{x^2}-\dfrac{x^2-y^2}{y^2}i\right)$?

Problems 21–26 involve concepts of infinite limits and limits at infinity discussed in (*i*) of the *Remarks*. In Problems 21–26, use (21) or (22), Theorem 3.1.2, and the basic limits (15) and (16) to compute the given complex limit.

**21.** $\lim\limits_{z\to\infty}\dfrac{z^2+iz-2}{(1+2i)z^2}$

**22.** $\lim\limits_{z\to\infty}\dfrac{iz+1}{2z-i}$

**23.** $\lim\limits_{z\to i}\dfrac{z^2-1}{z^2+1}$

**24.** $\lim\limits_{z\to -i/2}\dfrac{(1-i)z+i}{2z+i}$

**25.** $\lim\limits_{z\to\infty}\dfrac{z^2-(2+3i)z+1}{iz-3}$

**26.** $\lim\limits_{z\to i}\dfrac{z^2+1}{z^2+z+1-i}$

## 3.1.2 Continuity

In Problems 27–34, show that the function $f$ is continuous at the given point.

**27.** $f(z)=z^2-iz+3-2i;\ z_0=2-i$

**28.** $f(z)=z^3-\dfrac{1}{z};\ z_0=3i$

**29.** $f(z)=\dfrac{z^3}{z^3+3z^2+z};\ z_0=i$

**30.** $f(z)=\dfrac{z-3i}{z^2+2z-1};\ z_0=1+i$

**31.** $f(z)=\begin{cases}\dfrac{z^3-1}{z-1}, & |z|\neq 1\\ 3, & |z|=1\end{cases};\ z_0=1$

**32.** $f(z)=\begin{cases}\dfrac{z^3-1}{z^2+z+1}, & |z|\neq 1\\ \dfrac{-1+i\sqrt{3}}{2}, & |z|=1\end{cases};\ z_0=\dfrac{-1+i\sqrt{3}}{2}$

**33.** $f(z)=\bar{z}-3\,\mathrm{Re}(z)+i;\ z_0=3-2i$

**34.** $f(z)=\dfrac{\mathrm{Re}(z)}{z+iz}-2z^2;\ z_0=e^{i\pi/4}$

In Problems 35–40, show that the function $f$ is discontinuous at the given point.

**35.** $f(z)=\dfrac{z^2+1}{z+i};\ z_0=-i$

**36.** $f(z)=\dfrac{1}{|z|-1};\ z_0=i$

**37.** $f(z)=\mathrm{Arg}(z);\ z=-1$

**38.** $f(z)=\mathrm{Arg}(iz);\ z_0=i$

**39.** $f(z) = \begin{cases} \dfrac{z^3-1}{z-1}, & |z| \neq 1 \\ 3, & |z| = 1 \end{cases}$ ; $z_0 = i$

**40.** $f(z) = \begin{cases} \dfrac{z}{|z|}, & z \neq 0 \\ 1, & z = 0 \end{cases}$ ; $z_0 = 0$

In Problems 41–44, use Theorem 3.1.3 to determine the largest region in the complex plane on which the function $f$ is continuous.

**41.** $f(z) = \text{Re}(z)\,\text{Im}(z)$

**42.** $f(z) = \bar{z}$

**43.** $f(z) = \dfrac{z-1}{z\bar{z}-4}$

**44.** $f(z) = \dfrac{z^2}{(|z|-1)\,\text{Im}(z)}$

## Focus on Concepts

**45.** Use Theorem 3.1.1 to prove:

(**a**) $\lim\limits_{z\to z_0} c = c$, where $c$ is a constant.

(**b**) $\lim\limits_{z\to z_0} z = z_0$.

**46.** Use Theorem 3.1.1 to show that $\lim\limits_{z\to z_0} \bar{z} = \bar{z}_0$.

**47.** Use Theorem 3.1.2 and Problem 46 to show that

(**a**) $\lim\limits_{z\to z_0} \text{Re}(z) = \text{Re}(z_0)$.

(**b**) $\lim\limits_{z\to z_0} \text{Im}(z) = \text{Im}(z_0)$.

(**c**) $\lim\limits_{z\to z_0} |z| = |z_0|$.

**48.** Use Theorem 3.1.1 to prove part (*ii*) of Theorem 3.1.2.

**49.** The following is an epsilon-delta proof that $\lim\limits_{z\to z_0} z = z_0$. Fill in the missing parts.

**Proof** By Definition 3.1.1, $\lim\limits_{z\to z_0} z = z_0$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|_____| < \varepsilon$ whenever $0 < |_____| < \delta$. Setting $\delta =$ _____ will ensure that the previous statement is true.

**50.** The following is an epsilon-delta proof that $\lim\limits_{z\to z_0} \bar{z} = \bar{z}_0$. Provide the missing justifications in the proof.

**Proof** By Definition 3.1.1, $\lim\limits_{z\to z_0} \bar{z} = \bar{z}_0$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|_____| < \varepsilon$ whenever $0 < |_____| < \delta$. By properties of complex modulus and conjugation, $|z - z_0| = \left|\overline{z - z_0}\right| = |_____|$. Therefore, if $0 < |z - z_0| < \delta$ and $\delta =$ _____, then $|\bar{z} - \bar{z}_0| < \varepsilon$.

**51.** In this problem we will develop an epsilon-delta proof that $\lim\limits_{z\to 1+i} ((1-i)\,z + 2i) = 2 + 2i$.

(**a**) Write down the epsilon-delta definition (Definition 3.1.1) of $\lim\limits_{z\to 1+i} [(1-i)\,z + 2i] = 2 + 2i$.

(**b**) Factor out $(1-i)$ from the inequality involving $\varepsilon$ (from part (a)) and simplify. Now rewrite this inequality in the form $|z - (1+i)| <$ _____.

(**c**) Based on your work from part (b), what should $\delta$ be set equal to?

(**d**) Write an epsilon-delta proof that $\lim\limits_{z\to 1+i} [(1-i)\,z + 2i] = 2 + 2i$.

**52.** In this problem we will develop an epsilon-delta proof that $\lim\limits_{z\to 2i} \dfrac{2z^2 - 3iz + 2}{z - 2i} = 5i$.

(**a**) Write down the epsilon-delta definition (Definition 3.1.1) of $\lim\limits_{z\to 2i} \dfrac{2z^2 - 3iz + 2}{z - 2i} = 5i$.

(**b**) Simplify the inequality involving $\varepsilon$ (from part (a)), then rewrite this inequality in the form $|z - 2i| <$ _____.

(**c**) Based on your work from part (b), what should $\delta$ be set equal to?

(**d**) Write an epsilon-delta proof that $\lim\limits_{z\to 2i}\dfrac{2z^2-3iz+2}{z-2i}=5i$.

**53.** (**a**) Is it true that $\lim\limits_{z\to z_0}\overline{f(z)}=\lim\limits_{z\to z_0}f(\bar{z})$ for any complex function $f$? If so, then give a brief justification; if not, then find a counterexample.

(**b**) If $f(z)$ is a continuous function at $z_0$, then is it true that $\overline{f(z)}$ is continuous at $z_0$?

**54.** If $f$ is a function for which $\lim\limits_{x\to 0}f(x+i0)=0$ and $\lim\limits_{y\to 0}f(0+iy)=0$, then can you conclude that $\lim\limits_{z\to 0}f(z)=0$? Explain.

**55.** (**a**) Prove that the function $f(z)=\text{Arg}(z)$ is discontinuous at every point on the negative real axis.

(**b**) Prove that the function $f_1$ defined by

$$f_1(z)=\theta,\ -\pi<\theta<\pi$$

is a branch of the multiple-valued function $F(z)=\arg(z)$. [*Hint*: See Example 8.]

**56.** Consider the multiple-valued function $F(z)=z^{1/3}$ that assigns to $z$ the set of three cube roots of $z$. Explicitly define three distinct branches $f_1$, $f_2$, and $f_3$ of $F$, all of which have the nonnegative real axis as a branch cut.

**57.** Consider the multiple-valued function $F(z)=(z-1+i)^{1/2}$.

(**a**) What is the branch point of $F$? Explain.

(**b**) Explicitly define two distinct branches of $f_1$ and $f_2$ of $F$. In each case, state the branch cut.

**58.** Consider the multiple-valued function $F(z)=\left(z^2+1\right)^{1/2}$. What are the branch points (there are two of them) of $F$? Explain.

### Computer Lab Assignments

Reread part (*ii*) of the *Remarks* at the end of Section 3.1. In Problems 59–62, use a CAS to show that the given function is not continuous inside the unit circle by plotting the image of the given continuous parametric curve. (Be careful, *Mathematica* and *Maple* plots can sometimes be misleading.)

**59.** $f(z)=z+\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{2}\sqrt{3}it,\ -1\le t\le 1$

**60.** $f(z)=\sqrt[4]{r}e^{i\theta/4},\ \theta=\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{2}\sqrt{3}it,\ -1\le t\le 1$

**61.** $f(z)=\sqrt{r}e^{i\theta/2},\ \theta=\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{4}e^{it},\ 0\le t\le 2\pi$

**62.** $f(z)=|z-1|\text{Arg}(-z)+i\,\text{Arg}(iz),\ z(t)=\frac{1}{2}e^{it},\ 0\le t\le 2\pi$

## 3.2 Differentiability and Analyticity

**INTRODUCTION** The calculus of complex functions deals with the usual concepts of derivatives and integrals of these functions. In this section we shall give the limit definition of the derivative of a complex function $f(z)$. Although many of the concepts in this section will seem familiar to you, such as the product, quotient, and chain rules of differentiation, there are important differences between this material and the calculus of real functions $f(x)$. As the subsequent chapters of this text unfold, you will see that except for familiarity of names and definitions, there is little similarity between the *interpretations* of quantities such as $f'(x)$ and $f'(z)$.

## The Derivative

Suppose $z = x + iy$ and $z_0 = x_0 + iy_0$; then the change in $z_0$ is the difference $\Delta z = z - z_0$ or $\Delta z = x - x_0 + i(y - y_0) = \Delta x + i\Delta y$. If a complex function $w = f(z)$ is defined at $z$ and $z_0$, then the corresponding change in the function is the difference $\Delta w = f(z_0 + \Delta z) - f(z_0)$. The **derivative** of the function $f$ is defined in terms of a limit of the difference quotient $\Delta w/\Delta z$ as $\Delta z \to 0$.

### DEFINITION 3.2.1 Derivative of a Complex Function

Suppose the complex function $f$ is defined in a neighborhood of a point $z_0$. The **derivative** of $f$ at $z_0$, denoted by $f'(z_0)$, is

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \tag{1}$$

provided this limit exists.

If the limit in (1) exists, then the function $f$ is said to be **differentiable** at $z_0$. Two other symbols denoting the derivative of $w = f(z)$ are $w'$ and $dw/dz$. If the latter notation is used, then the value of a derivative at a specified point $z_0$ is written $\left.\dfrac{dw}{dz}\right|_{z=z_0}$.

### EXAMPLE 1 Using Definition 3.2.1

Use Definition 3.2.1 to find the derivative of $f(z) = z^2 - 5z$.

**Solution** Because we are going to compute the derivative of $f$ at any point, we replace $z_0$ in (1) by the symbol $z$. First,

$$f(z + \Delta z) = (z + \Delta z)^2 - 5(z + \Delta z) = z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z.$$

Second,

$$\begin{aligned} f(z + \Delta z) - f(z) &= z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z - (z^2 - 5z) \\ &= 2z\Delta z + (\Delta z)^2 - 5\Delta z. \end{aligned}$$

Then, finally, (1) gives

$$\begin{aligned} f'(z) &= \lim_{\Delta z \to 0} \frac{2z\Delta z + (\Delta z)^2 - 5\Delta z}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{\Delta z(2z + \Delta z - 5)}{\Delta z} \\ &= \lim_{\Delta z \to 0} (2z + \Delta z - 5). \end{aligned}$$

The limit is $f'(z) = 2z - 5$. ☰

## Rules of Differentiation

The familiar rules of differentiation in the calculus of real variables carry over to the calculus of complex variables. If $f$ and $g$ are differentiable at a point $z$, and $c$ is a complex constant, then (1) can be used to show:

## Differentiation Rules

*Constant Rules*: $\dfrac{d}{dz}c = 0$ and $\dfrac{d}{dz}cf(z) = cf'(z)$ (2)

*Sum Rule*: $\dfrac{d}{dz}[f(z) \pm g(z)] = f'(z) \pm g'(z)$ (3)

*Product Rule*: $\dfrac{d}{dz}[f(z)g(z)] = f(z)g'(z) + f'(z)g(z)$ (4)

*Quotient Rule*: $\dfrac{d}{dz}\left[\dfrac{f(z)}{g(z)}\right] = \dfrac{g(z)f'(z) - f(z)g'(z)}{[g(z)]^2}$ (5)

*Chain Rule*: $\dfrac{d}{dz}f(g(z)) = f'(g(z))\,g'(z).$ (6)

The **power rule** for differentiation of powers of $z$ is also valid:

$$\frac{d}{dz}z^n = nz^{n-1}, \quad n \text{ an integer.} \tag{7}$$

Combining (7) with (6) gives the **power rule for functions**:

$$\frac{d}{dz}[g(z)]^n = n[g(z)]^{n-1}g'(z), \quad n \text{ an integer.} \tag{8}$$

### EXAMPLE 2 Using the Rules of Differentiation

Differentiate:

**(a)** $f(z) = 3z^4 - 5z^3 + 2z$ **(b)** $f(z) = \dfrac{z^2}{4z+1}$ **(c)** $f(z) = \left(iz^2 + 3z\right)^5$

**Solution** **(a)** Using the power rule (7), the sum rule (3), along with (2), we obtain

$$f'(z) = 3 \cdot 4z^3 - 5 \cdot 3z^2 + 2 \cdot 1 = 12z^3 - 15z^2 + 2.$$

**(b)** From the quotient rule (5),

$$f'(z) = \frac{(4z+1)\cdot 2z - z^2 \cdot 4}{(4z+1)^2} = \frac{4z^2 + 2z}{(4z+1)^2}.$$

**(c)** In the power rule for functions (8) we identify $n = 5$, $g(z) = iz^2 + 3z$, and $g'(z) = 2iz + 3$, so that

$$f'(z) = 5(iz^2 + 3z)^4(2iz + 3).$$

For a complex function $f$ to be differentiable at a point $z_0$, we know from the preceding section that the limit $\lim\limits_{\Delta z \to 0} \dfrac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ must exist and equal the same complex number from any direction; that is, the limit must exist regardless of how $\Delta z$ approaches 0. This means that in complex analysis, the requirement of differentiability of a function $f(z)$ at a point $z_0$ is a far greater demand than in real calculus of functions $f(x)$ where we can approach a real number $x_0$ on the number line from only two directions. If a complex function is made up by specifying its real

and imaginary parts $u$ and $v$, such as $f(z) = x + 4iy$, there is a good chance that it is not differentiable.

### EXAMPLE 3 A Function That Is Nowhere Differentiable

Show that the function $f(z) = x + 4iy$ is not differentiable at any point $z$.

**Solution** Let $z$ be any point in the complex plane. With $\Delta z = \Delta x + i\,\Delta y$,

$$f(z + \Delta z) - f(z) = (x + \Delta x) + 4i(y + \Delta y) - x - 4iy = \Delta x + 4i\,\Delta y$$

and so
$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x + 4i\,\Delta y}{\Delta x + i\,\Delta y}. \tag{9}$$

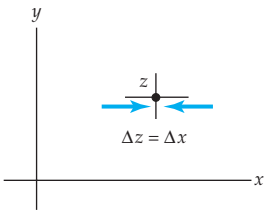
Now, as shown in FIGURE 3.2.1(a), if we let $\Delta z \to 0$ along a line parallel to the $x$-axis, then $\Delta y = 0$ and $\Delta z = \Delta x$. Along this path we have:

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x}{\Delta x} = 1. \tag{10}$$

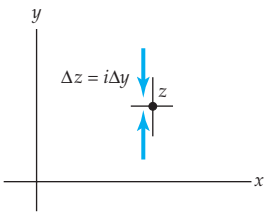
On the other hand, if we let $\Delta z \to 0$ along a line parallel to the $y$-axis as shown in Figure 3.2.1(b), then $\Delta x = 0$ and $\Delta z = i\,\Delta y$ so that

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{4i\,\Delta y}{i\,\Delta y} = 4. \tag{11}$$

In view of the obvious fact that the values in (10) and (11) are different, we conclude that the limit in (9) does not exist. Therefore, $f(z) = x + 4iy$ is nowhere differentiable; that is, $f$ is not differentiable at any point $z$. ☰




**(a)** $\Delta z \to 0$ along a line parallel to $x$-axis




**(b)** $\Delta z \to 0$ along a line parallel to $y$-axis

**FIGURE 3.2.1** Approaching $z$ along a horizontal line and then along a vertical line

The basic power rule (7) does not apply to powers of the conjugate of $z$ because, like the function in Example 3, the function $f(z) = \bar{z}$ is nowhere differentiable. See Problem 21 in Exercises 3.2.

**Analytic Functions** Even though the requirement of differentiability is a stringent demand, there is a class of functions that is of great importance whose members satisfy even more severe requirements. These functions are called **analytic functions**.

### DEFINITION 3.2.2 Analyticity at a Point

A complex function $w = f(z)$ is said to be **analytic at a point** $z_0$ if $f$ is differentiable at $z_0$ and at every point in some neighborhood of $z_0$.

A function $f$ **is analytic in a domain** $D$ if it is analytic at every point in $D$. The phrase "analytic *on* a domain $D$" is also used. Although we shall not use these terms in this text, a function $f$ that is analytic throughout a domain $D$ is called **holomorphic** or **regular**.

*Very Important* ▶ You should reread Definition 3.2.2 carefully. Analyticity at a point is *not* the same as differentiability at a point. Analyticity *at a point is a neighborhood property*; in other words, analyticity is a property that is defined over an open set. It is left as an exercise to show that the function $f(z) = |z|^2$ is differentiable at $z = 0$ but is not differentiable anywhere else. Even though $f(z) = |z|^2$ is differentiable at $z = 0$, it is not analytic at that point because there exists no neighborhood of $z = 0$ *throughout which* $f$ is differentiable; hence the function $f(z) = |z|^2$ is nowhere analytic. See Problem 19 in Exercises 3.2.

In contrast, the simple polynomial $f(z) = z^2$ is differentiable at every point $z$ in the complex plane. Hence, $f(z) = z^2$ is analytic everywhere.

**Entire Functions** A function that is analytic at every point $z$ in the complex plane is said to be an **entire function**. In view of differentiation rules (2), (3), (7), and (5), we can conclude that polynomial functions are differentiable at every point $z$ in the complex plane and rational functions are analytic throughout any domain $D$ that contains no points at which the denominator is zero. The following theorem summarizes these results.

**THEOREM 3.2.1 Polynomial and Rational Functions**

(*i*) A complex polynomial function $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, where $n$ is a nonnegative integer, is an entire function.

(*ii*) A complex rational function $f(z) = \dfrac{p(z)}{q(z)}$, where $p$ and $q$ are polynomial functions, is analytic in any domain $D$ that contains no point $z_0$ for which $q(z_0) = 0$.

**Singular Points** Since the rational function $f(z) = 4z/\left(z^2 - 2z + 2\right)$ is undefined at $1 + i$ and $1 - i$, $f$ fails to be analytic at these points. Thus by (*ii*) of Theorem 3.2.1, $f$ is not analytic in any domain containing one or both of these points. In general, a point $z$ at which a complex function $w = f(z)$ fails to be analytic is called a **singular point** of $f$. We will discuss singular points in greater depth in Chapter 6.

$1 \pm i$ are zeros of the denominator of $f$. ▶

Another consequence of the differentiation rules on page 123 is:

**Analyticity of Sum, Product, and Quotient**

*If the functions $f$ and $g$ are analytic in a domain $D$, then the sum $f(z) + g(z)$, difference $f(z) - g(z)$, and product $f(z)g(z)$ are analytic in $D$. The quotient $f(z)/g(z)$ is analytic provided $g(z) \neq 0$ in $D$.*

**An Alternative Definition of $f'(z)$** Sometimes it is convenient to define the derivative of a function $f$ using an alternative form of the difference quotient $\Delta w/\Delta z$. Since $\Delta z = z - z_0$, then $z = z_0 + \Delta z$, and so (1) can be written as

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}. \tag{12}$$

In contrast to what we did in Example 1, if we wish to compute $f'$ at a general point $z$ using (12), then we replace $z_0$ by the symbol $z$ *after* the limit is computed. See Problems 7–10 in Exercises 3.2.

As in real analysis, if a complex function $f$ is differentiable at a point, then the function is necessarily continuous at the point. We use the form of the derivative given in (12) to prove the last statement.

**THEOREM 3.2.2 Differentiability Implies Continuity**

If $f$ is differentiable at a point $z_0$ in a domain $D$, then $f$ is continuous at $z_0$.

**Proof** The limits $\lim_{z\to z_0} \dfrac{f(z)-f(z_0)}{z-z_0}$ and $\lim_{z\to z_0}(z-z_0)$ exist and equal $f'(z_0)$ and 0, respectively. Hence by Theorem 3.1.2(*iii*), we can write the following limit of a product as the product of the limits:

$$\begin{aligned}\lim_{z\to z_0}(f(z)-f(z_0)) &= \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\cdot(z-z_0)\\ &= \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\cdot\lim_{z\to z_0}(z-z_0) = f'(z_0)\cdot 0 = 0.\end{aligned}$$

From $\lim_{z\to z_0}(f(z)-f(z_0)) = 0$ we conclude that $\lim_{z\to z_0} f(z) = f(z_0)$. In view of Definition 3.1.2, $f$ is continuous at $z_0$. ☰

Of course the converse of Theorem 3.2.2 is not true; continuity of a function $f$ at a point does not guarantee that $f$ is differentiable at the point. It follows from Theorem 3.1.3 that the simple function $f(z) = x + 4iy$ is continuous everywhere because the real and imaginary parts of $f$, $u(x, y) = x$ and $v(x, y) = 4y$ are continuous at any point $(x, y)$. Yet we saw in Example 3 that $f(z) = x + 4iy$ is not differentiable at any point $z$.

As another consequence of differentiability, L'Hôpital's rule for computing limits of the indeterminate form $0/0$, carries over to complex analysis.

**THEOREM 3.2.3 L'Hôpital's Rule**

Suppose $f$ and $g$ are functions that are analytic at a point $z_0$ and $f(z_0) = 0$, $g(z_0) = 0$, but $g'(z_0) \neq 0$. Then

$$\lim_{z\to z_0}\frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}. \tag{13}$$

The task of establishing (13) is neither long nor difficult. You are guided through the steps of a proof in Problem 33 in Exercises 3.2.

**EXAMPLE 4 Using L'Hôpital's Rule**

Compute $\lim_{z\to 2+i}\dfrac{z^2-4z+5}{z^3-z-10i}$.

**Solution** If we identify $f(z) = z^2 - 4z + 5$ and $g(z) = z^3 - z - 10i$, you should verify that $f(2+i) = 0$ and $g(2+i) = 0$. The given limit has the indeterminate form $0/0$. Now since $f$ and $g$ are polynomial functions, both functions are necessarily analytic at $z_0 = 2 + i$. Using

$$f'(z) = 2z - 4, \quad g'(z) = 3z^2 - 1, \quad f'(2+i) = 2i, \quad g'(2+i) = 8 + 12i,$$

we see that (13) gives

$$\lim_{z\to 2+i}\frac{z^2-4z+5}{z^3-z-10i} = \frac{f'(2+i)}{g'(2+i)} = \frac{2i}{8+12i} = \frac{3}{26}+\frac{1}{13}i.$$ ☰

In part (b) of Example 4 in Section 3.1 we resorted to the lengthy procedure of factoring and cancellation to compute the limit

$$\lim_{z\to 1+\sqrt{3}i} \frac{z^2-2z+4}{z-1-\sqrt{3}i}. \tag{14}$$

A rereading of that example shows that the limit (14) has the indeterminate form $0/0$. With $f(z) = z^2 - 2z + 4$, $g(z) = z - 1 - \sqrt{3}i$, $f'(z) = 2z - 2$, and $g'(z) = 1$, L'Hôpital's rule (13) gives immediately

$$\lim_{z\to 1+\sqrt{3}i} \frac{z^2-2z+4}{z-1-\sqrt{3}i} = \frac{f'(1+\sqrt{3}i)}{1} = 2\left(1+\sqrt{3}i-1\right) = 2\sqrt{3}i.$$

## *Remarks* COMPARISON WITH REAL ANALYSIS

(*i*) In real calculus the derivative of a function $y = f(x)$ at a point $x$ has many interpretations. For example, $f'(x)$ is the slope of the tangent line to the graph of $f$ at the point $(x,\ f(x))$. When the slope is positive, negative, or zero, the function, in turn, is increasing, decreasing, and possibly has a maximum or minimum. Also, $f'(x)$ is the instantaneous rate of change of $f$ at $x$. In a physical setting, this rate of change can be interpreted as velocity of a moving object. *None of these interpretations carry over to complex calculus*. Thus it is fair to ask: What does the derivative of a complex function $w = f(z)$ represent? Here is the answer: In complex analysis the primary concern is *not* what a derivative of function *is* or *represents*, but rather, it is whether a function $f$ actually *has* a derivative. The fact that a complex function $f$ possesses a derivative tells us a lot about the function. As we have just seen, when $f$ is differentiable at $z$ and at every point in some neighborhood of $z$, then $f$ is analytic at the point $z$. You will see the importance of analytic functions in the remaining chapters of this book. For example, the derivative plays an important role in the theory of mappings by complex functions. Roughly, under a mapping defined by an analytic function $f$, the magnitude and sense of an angle between two curves that intersect a point $z_0$ in the $z$-plane is preserved in the $w$-plane at all points at which $f'(z) \neq 0$. See Chapter 7.

(*ii*) We pointed out in the foregoing discussion that $f(z) = |z|^2$ was differentiable only at the single point $z = 0$. In contrast, the real function $f(x) = |x|^2$ is differentiable everywhere. The real function $f(x) = x$ is differentiable everywhere, but the complex function $f(z) = x = \text{Re}(z)$ is nowhere differentiable.

(*iii*) The differentiation formulas (2)–(8) are important, but not nearly as important as in real analysis. In complex analysis we deal with functions such as $f(z) = 4x^2 - iy$ and $g(z) = xy + i(x + y)$, which, even if they possess derivatives, cannot be differentiated by formulas (2)–(8).

(*iv*) In this section we have not mentioned the concept of higher-order derivatives of complex functions. We will pursue this topic in depth in Section 5.5. There is nothing surprising about the definitions of higher derivatives; they are defined in exactly the same manner as in real analysis. For example, the second derivative is the derivative of the first derivative. In the case $f(z) = 4z^3$ we see that $f'(z) = 12z^2$ and so the second derivative is $f''(z) = 24z$. But there is a major difference between real and complex

variables concerning the existence of higher-order derivatives. In real analysis, if a function $f$ possesses, say, a first derivative, there is no guarantee that $f$ possesses any other higher derivatives. For example, on the interval $(-1,\ 1)$, $f(x) = x^{3/2}$ is differentiable at $x = 0$, but $f'(x) = \frac{3}{2}x^{1/2}$ is not differentiable at $x = 0$. In complex analysis, if a function $f$ is *analytic* in a domain $D$ then, by assumption, $f$ possesses a derivative at each point in $D$. We will see in Section 5.5 that this fact alone guarantees that $f$ possesses higher-order derivatives at all points in $D$. Indeed, an analytic function $f$ is infinitely differentiable in $D$.

(*v*) The definition of "analytic at a point $a$" in real analysis differs from the usual definition of that concept in complex analysis (Definition 3.2.2). In real analysis, analyticity of a function is defined in terms of power series: A function $y = f(x)$ is analytic at a point $a$ if $f$ has a Taylor series at $a$ that represents $f$ in some neighborhood of $a$. In view of *Remark* (*iv*), why are these two definitions not really that different?

(*vi*) As in real calculus, it may be necessary to apply L'Hôpital's rule several times in succession to calculate a limit. In other words, if $f(z_0),\ g(z_0),\ f'(z_0)$, and $g'(z_0)$ are all zero, the limit $\lim\limits_{z\to z_0} f(z)/g\,(z)$ may still exist. In general, if $f$, $g$, and their first $n-1$ derivatives are zero at $z_0$ and $g^{(n)}(z_0) \neq 0$, then

$$\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{f^{(n)}(z_0)}{g^{(n)}(z_0)}.$$

## EXERCISES 3.2

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1–6, use (1) of Definition 3.2.1 to find $f'(z)$ for the given function.

**1.** $f(z) = 9iz + 2 - 3i$

**2.** $f(z) = 15z^2 - 4z + 1 - 3i$

**3.** $f(z) = iz^3 - 7z^2$

**4.** $f(z) = \dfrac{1}{z}$

**5.** $f(z) = z - \dfrac{1}{z}$

**6.** $f(z) = -z^{-2}$

In Problems 7–10, use the alternative definition (12) to find $f'(z)$ for the given function.

**7.** $f(z) = 5z^2 - 10z + 8$

**8.** $f(z) = z^3$

**9.** $f(z) = z^4 - z^2$

**10.** $f(z) = \dfrac{1}{2iz}$

In Problems 11–18, use the rules of differentiation to find $f'(z)$ for the given function.

**11.** $f(z) = (2-i)z^5 + iz^4 - 3z^2 + i^6$

**12.** $f(z) = 5(iz)^3 - 10z^2 + 3 - 4i$

**13.** $f(z) = (z^6 - 1)(z^2 - z + 1 - 5i)$

**14.** $f(z) = (z^2 + 2z - 7i)^2(z^4 - 4iz)^3$

**15.** $f(z) = \dfrac{iz^2 - 2z}{3z + 1 - i}$

**16.** $f(z) = -5iz^2 + \dfrac{2+i}{z^2}$

**17.** $f(z) = \left(z^4 - 2iz^2 + z\right)^{10}$

**18.** $f(z) = \left(\dfrac{(4+2i)z}{(2-i)z^2 + 9i}\right)^3$

**19.** The function $f(z) = |z|^2$ is continuous at the origin.

(**a**) Show that $f$ *is* differentiable at the origin.

(**b**) Show that $f$ *is not* differentiable at any point $z \neq 0$.

**20.** Show that the function

$$f(z) = \begin{cases} 0, & z = 0 \\ \dfrac{x^3 - y^3}{x^2 + y^2} + i\dfrac{x^3 + y^3}{x^2 + y^2}, & z \neq 0 \end{cases}$$

is not differentiable at $z = 0$ by letting $\Delta z \to 0$ first along the $x$-axis and then along the line $y = x$.

In Problems 21 and 22, proceed as in Example 3 to show that the given function is nowhere differentiable.

**21.** $f(z) = \bar{z}$

**22.** $f(z) = |z|$

In Problems 23–26, use L'Hôpital's rule to compute the given limit.

**23.** $\lim\limits_{z \to i} \dfrac{z^7 + i}{z^{14} + 1}$

**24.** $\lim\limits_{z \to \sqrt{2}+\sqrt{2}i} \dfrac{z^4 + 16}{z^2 - 2\sqrt{2}z + 4}$

**25.** $\lim\limits_{z \to 1+i} \dfrac{z^5 + 4z}{z^2 - 2z + 2}$

**26.** $\lim\limits_{z \to \sqrt{2}i} z\dfrac{z^3 + 5z^2 + 2z + 10}{z^5 + 2z^3}$

In Problems 27–30, determine the points at which the given function is not analytic.

**27.** $f(z) = \dfrac{iz^2 - 2z}{3z + 1 - i}$

**28.** $f(z) = -5iz^2 + \dfrac{2 + i}{z^2}$

**29.** $f(z) = \left(z^4 - 2iz^2 + z\right)^{10}$

**30.** $f(z) = \left(\dfrac{(2 - i)z + 1}{z^2 + 9i}\right)^3$

## Focus on Concepts

**31.** Use Definition 3.2.1 and Theorem 3.1.2 to prove the constant rule $\dfrac{d}{dz}cf(z) = cf'(z)$.

**32.** Use Definition 3.2.1 and Theorem 3.1.2 to prove the sum rule $\dfrac{d}{dz}[f(z) + g(z)] = f'(z) + g'(z)$.

**33.** In this problem you are guided through the start of the proof of the proposition:

*If functions $f$ and $g$ are analytic at a point $z_0$ and $f(z_0) = 0,\ g(z_0) = 0$, but $g'(z_0) \neq 0$, then* $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)} = \dfrac{f'(z_0)}{g'(z_0)}$.

**Proof** We begin with the hypothesis that $f$ and $g$ are analytic at a point $z_0$. Analyticity at $z_0$ implies $f$ and $g$ are differentiable at $z_0$. Hence from (12) both limits,

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad \text{and} \quad g'(z_0) = \lim_{z \to z_0} \frac{g(z) - g(z_0)}{z - z_0}$$

exist. But since $f(z_0) = 0,\ g(z_0) = 0$, the foregoing limits are the same as

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z)}{z - z_0} \quad \text{and} \quad g'(z_0) = \lim_{z \to z_0} \frac{g(z)}{z - z_0}.$$

Now examine $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)}$ and finish the proof.

**34.** In this problem you are guided through the start of the proof of the product rule.

**Proof** We begin with the hypothesis that $f$ and $g$ are differentiable at a point $z$; that is, each of the following limits exist:

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad \text{and} \quad g'(z) = \lim_{\Delta z \to 0} \frac{g(z + \Delta z) - g(z)}{\Delta z}$$

(**a**) Justify the equality

$$\frac{d}{dz}[f(z)g(z)] = \lim_{\Delta z \to 0} \frac{f(z+\Delta z)g(z+\Delta z) - f(z)g(z)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \left[ \frac{f(z+\Delta z) - f(z)}{\Delta z} g(z+\Delta z) + f(z) \frac{g(z+\Delta z) - g(z)}{\Delta z} \right].$$

(**b**) Use Definition 3.1.2 to justify $\lim_{\Delta z \to 0} g(z+\Delta z) = g(z)$.

(**c**) Use Theorems 3.1.2(*ii*) and 3.1.2(*iii*) to finish the proof.

**35.** In this problem you will prove that $f(z) = \overline{z}$ is nowhere differentiable using the polar form $\Delta z = r(\cos\theta + i\sin\theta)$.

(**a**) What must $r$ approach if $\Delta z \to 0$? Must $\theta$ approach a specific value?

(**b**) If $f(z) = \overline{z}$, show that

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{\cos\theta - i\sin\theta}{\cos\theta + i\sin\theta}.$$

(**c**) Explain succinctly why the result in part (b) shows that $f$ is nowhere differentiable.

**36.** In this problem you will prove that $f(z) = |z|$ is nowhere differentiable using the alternative definition (12) and polar forms of $z$ and $z_0$.

(**a**) Let $z_0 = r(\cos\theta + i\sin\theta)$ and $z = (r+\Delta r)(\cos\theta + i\sin\theta)$. Along this path determine $\lim_{z \to z_0} \dfrac{f(z) - f(z_0)}{z - z_0}$.

(**b**) Let $z_0 = r(\cos\theta + i\sin\theta)$ and $z = r\,[\cos(\theta + \Delta\theta) + i\sin(\theta + \Delta\theta)]$. Along this path determine $\lim_{z \to z_0} \dfrac{f(z) - f(z_0)}{z - z_0}$.

(**c**) Explain succinctly why the results in parts (a) and (b) show that $f$ is nowhere differentiable.

**37.** Suppose $f'(z)$ exists at a point $z$. Is $f'(z)$ continuous at $z$?

**38.** (**a**) Let $f(z) = z^2$. Write down the real and imaginary parts of $f$ and $f'$. What do you observe?

(**b**) Repeat part (a) for $f(z) = 3iz + 2$.

(**c**) Make a conjecture about the relationship between real and imaginary parts of $f$ versus $f'$.

## 3.3 Cauchy-Riemann Equations

**INTRODUCTION** In the preceding section we saw that a function $f$ of a complex variable $z$ is *analytic at a point* $z$ when $f$ is differentiable at $z$ and differentiable at every point in some neighborhood of $z$. This requirement is more stringent than simply *differentiability at a point* because a complex function can be differentiable at a point $z$ but yet be differentiable nowhere else. A function $f$ is *analytic in a domain* $D$ if $f$ is differentiable at all points in $D$. We shall now develop a test for analyticity of a complex function $f(z) = u(x, y) + iv(x, y)$ that is based on partial derivatives of its real and imaginary parts $u$ and $v$.

### A Necessary Condition for Analyticity

A review of partial derivatives can be found in Section 3.3 of the student study guide. ▶

In the next theorem we see that if a function $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point $z$, then the functions $u$ and $v$ must satisfy a pair of equations that relate their first-order partial derivatives.

**THEOREM 3.3.1 Cauchy-Riemann Equations**

Suppose $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point $z = x + iy$. Then at $z$ the first-order partial derivatives of $u$ and $v$ exist and satisfy the **Cauchy-Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{1}$$

**Proof** The derivative of $f$ at $z$ is given by

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}. \tag{2}$$

By writing $f(z) = u(x, y) + iv(x, y)$ and $\Delta z = \Delta x + i\Delta y$, (2) becomes

$$f'(z) = \lim_{\Delta z \to 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}. \tag{3}$$

Since the limit (2) is assumed to exist, $\Delta z$ can approach zero from any convenient direction. In particular, if we choose to let $\Delta z \to 0$ along a horizontal line, then $\Delta y = 0$ and $\Delta z = \Delta x$. We can then write (3) as

$$\begin{aligned} f'(z) &= \lim_{\Delta x \to 0} \frac{u(x + \Delta x, y) - u(x, y) + i\,[v(x + \Delta x, y) - v(x, y)]}{\Delta x} \\ &= \lim_{\Delta x \to 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}. \end{aligned} \tag{4}$$

The existence of $f'(z)$ implies that each limit in (4) exists. These limits are the definitions of the first-order partial derivatives with respect to $x$ of $u$ and $v$, respectively. Hence, we have shown two things: both $\partial u/\partial x$ and $\partial v/\partial x$ exist at the point $z$, and that the derivative of $f$ is

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}. \tag{5}$$

We now let $\Delta z \to 0$ along a vertical line. With $\Delta x = 0$ and $\Delta z = i\Delta y$, (3) becomes

$$f'(z) = \lim_{\Delta y \to 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \to 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y}. \tag{6}$$

In this case (6) shows us that $\partial u/\partial y$ and $\partial v/\partial y$ exist at $z$ and that

$$f'(z) = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \tag{7}$$

By equating the real and imaginary parts of (5) and (7) we obtain the pair of equations in (1). ☰

Because Theorem 3.3.1 states that the Cauchy-Riemann equations (1) hold at $z$ as a *necessary* consequence of $f$ being differentiable at $z$, we cannot use the theorem to help us determine where $f$ is differentiable. *But* it is important to realize that Theorem 3.3.1 can tell us where a function $f$ does not possess a derivative. If the equations in (1) are *not* satisfied at a point $z$, then $f$ cannot be differentiable at $z$. We have already seen in Example 3 of Section 3.2 that $f(z) = x + 4iy$ is not differentiable at any

point $z$. If we identify $u = x$ and $v = 4y$, then $\partial u/\partial x = 1$, $\partial v/\partial y = 4$, $\partial u/\partial y = 0$, and $\partial v/\partial x = 0$. In view of

$$\frac{\partial u}{\partial x} = 1 \neq \frac{\partial v}{\partial y} = 4$$

the two equations in (1) cannot be simultaneously satisfied at any point $z$. Therefore, by Theorem 3.3.1, $f$ is nowhere differentiable.

It also follows from Theorem 3.3.1 that if a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic throughout a domain $D$, then the real functions $u$ and $v$ satisfy the Cauchy-Riemann equations (1) at every point in $D$.

### EXAMPLE 1 Verifying Theorem 3.3.1

The polynomial function $f(z) = z^2 + z$ is analytic for all $z$ and can be written as $f(z) = x^2 - y^2 + x + i(2xy + y)$. Thus, $u(x, y) = x^2 - y^2 + x$ and $v(x, y) = 2xy + y$. For any point $(x, y)$ in the complex plane we see that the Cauchy-Riemann equations are satisfied:

$$\frac{\partial u}{\partial x} = 2x + 1 = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}.$$

The contrapositive* form of the sentence preceding Example 1 is:

**Criterion for Non-analyticity**

*If the Cauchy-Riemann equations are not satisfied at every point $z$ in a domain $D$, then the function $f(z) = u(x, y) + iv(x, y)$ cannot be analytic in $D$.*

### EXAMPLE 2 Using the Cauchy-Riemann Equations

Show that the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ is not analytic at any point.

**Solution** We identify $u(x, y) = 2x^2 + y$ and $v(x, y) = y^2 - x$. From

$$\begin{aligned} \frac{\partial u}{\partial x} &= 4x \quad \text{and} \quad \frac{\partial v}{\partial y} = 2y \\ \frac{\partial u}{\partial y} &= 1 \quad \text{and} \quad \frac{\partial v}{\partial x} = -1 \end{aligned} \tag{8}$$

we see that $\partial u/\partial y = -\partial v/\partial x$ but that the equality $\partial u/\partial x = \partial v/\partial y$ is satisfied only on the line $y = 2x$. However, for any point $z$ on the line, there is no neighborhood or open disk about $z$ in which the Cauchy-Riemann equations are satisfied. We conclude that $f$ is nowhere analytic.

**A Sufficient Condition for Analyticity** By themselves, the Cauchy-Riemann equations do not ensure analyticity of a function $f(z) = u(x, y) + iv(x, y)$ at a point $z = x + iy$. It is possible for the Cauchy-Riemann equations to be satisfied at $z$ and yet $f(z)$ may not be differentiable at $z$, or $f(z)$ may be differentiable at $z$ but nowhere else. In either case, $f$ is not analytic at $z$. See Problem 40 in Exercises 3.3. However, when we add the condition of continuity to $u$ and $v$ *and* to

*A proposition "If $P$, then $Q$" is logically equivalent to its contrapositive "If not $Q$, then not $P$."

the four partial derivatives $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, and $\partial v/\partial y$, it can be shown that the Cauchy-Riemann equations are not only necessary but also sufficient to guarantee analyticity of $f(z) = u(x, y) + iv(x, y)$ at $z$. The proof is long and complicated and so we state only the result.

**THEOREM 3.3.2 Criterion for Analyticity**

Suppose the real functions $u(x, y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in a domain $D$. If $u$ and $v$ satisfy the Cauchy-Riemann equations (1) at all points of $D$, then the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in $D$.

## EXAMPLE 3 Using Theorem 3.3.2

For the function $f(z) = \dfrac{x}{x^2+y^2} - i\dfrac{y}{x^2+y^2}$, the real functions $u(x, y) = \dfrac{x}{x^2+y^2}$ and $v(x, y) = -\dfrac{y}{x^2+y^2}$ are continuous except at the point where $x^2 + y^2 = 0$, that is, at $z = 0$. Moreover, the four first-order partial derivatives

$$\frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2+y^2)^2}, \qquad \frac{\partial u}{\partial y} = -\frac{2xy}{(x^2+y^2)^2},$$
$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2+y^2)^2}, \quad \text{and} \quad \frac{\partial v}{\partial y} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

are continuous except at $z = 0$. Finally, we see from

$$\frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2+y^2)^2} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{2xy}{(x^2+y^2)^2} = -\frac{\partial v}{\partial x}$$

that the Cauchy-Riemann equations are satisfied except at $z = 0$. Thus we conclude from Theorem 3.3.2 that $f$ is analytic in any domain $D$ that does not contain the point $z = 0$. ☰

The results in (5) and (7) were obtained under the basic assumption that $f$ was differentiable at the point $z$. In other words, if $f$ is differentiable, then (5) and (7) give us a formula for computing the derivative $f'(z)$:

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}. \tag{9}$$

For example, we already know from part $(i)$ of Theorem 3.2.1 that $f(z) = z^2$ is entire and so is differentiable for all $z$. With $u(x, y) = x^2 - y^2$, $\partial u/\partial x = 2x$, $v(x, y) = 2xy$, and $\partial v/\partial y = 2y$, we see from (9) that

$$f'(z) = 2x + i2y = 2(x + iy) = 2z.$$

For Example 3, the function $f(x) = \dfrac{x}{x^2+y^2} - i\dfrac{y}{x^2+y^2}$ has derivative

$$f'(z) = \frac{y^2 - x^2}{(x^2+y^2)^2} + i\frac{2xy}{(x^2+y^2)^2}$$

in any domain that does not include $z = 0$.

Recall that analyticity implies differentiability but not conversely. Theorem 3.3.2 has an analogue that gives the following criterion for differentiability.

### Sufficient Conditions for Differentiability

*If the real functions $u(x, y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in some neighborhood of a point $z$, and if $u$ and $v$ satisfy the Cauchy-Riemann equations* (1) *at $z$, then the complex function $f(z) = u(x, y) + iv(x, y)$ is differentiable at $z$ and $f'(z)$ is given by* (9).

**EXAMPLE 4 A Function Differentiable on a Line**

In Example 2 we saw that the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ was nowhere analytic, but yet the Cauchy-Riemann equations were satisfied on the line $y = 2x$. Since the functions $u(x, y) = 2x^2 + y$, $\partial u/\partial x = 4x$, $\partial u/\partial y = 1$, $v(x, y) = y^2 - x$, $\partial v/\partial x = -1$, and $\partial v/\partial y = 2y$ are continuous at every point, it follows that $f$ is differentiable on the line $y = 2x$. Moreover, from (9) we see that the derivative of $f$ at points on this line is given by $f'(z) = 4x - i = 2y - i$. ☰

The following theorem is a direct consequence of the Cauchy-Riemann equations. Its proof is left as an exercise. See Problems 31 and 35 in Exercises 3.3.

**THEOREM 3.3.3 Constant Functions**

Suppose the function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain $D$.

(*i*) If $|f(z)|$ is constant in $D$, then so is $f(z)$.

(*ii*) If $f'(z) = 0$ in $D$, then $f(z) = c$ in $D$, where $c$ is a complex constant.

**Polar Coordinates** In Section 2.1 we saw that a complex function can be expressed in terms of polar coordinates. Indeed, the form $f(z) = u(r, \theta) + iv(r, \theta)$ is often more convenient to use. In polar coordinates the Cauchy-Riemann equations become

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}. \tag{10}$$

The polar version of (9) at a point $z$ whose polar coordinates are $(r, \theta)$ is then

$$f'(z) = e^{-i\theta}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right) = \frac{1}{r}e^{-i\theta}\left(\frac{\partial v}{\partial \theta} - i\frac{\partial u}{\partial \theta}\right). \tag{11}$$

See Problems 38 and 39 in Exercises 3.3.

### *Remarks* COMPARISON WITH REAL ANALYSIS

In real calculus, one of the noteworthy properties of the exponential function $f(x) = e^x$ is that $f'(x) = e^x$. In (3) of Section 2.1 we gave the definition of the complex exponential $f(z) = e^z$. We are now in a position to show that $f(z) = e^z$ is differentiable everywhere and that this complex function shares the same derivative property as its real counterpart, that is, $f'(z) = f(z)$. See Problem 25 in Exercises 3.3.

## EXERCISES 3.3

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1 and 2, the given function is analytic for all $z$. Show that the Cauchy-Riemann equations are satisfied at every point.

**1.** $f(z) = z^3$

**2.** $f(z) = 3z^2 + 5z - 6i$

In Problems 3–8, show that the given function is not analytic at any point.

**3.** $f(z) = \text{Re}(z)$

**4.** $f(z) = y + ix$

**5.** $f(z) = 4z - 6\bar{z} + 3$

**6.** $f(z) = \bar{z}^2$

**7.** $f(z) = x^2 + y^2$

**8.** $f(z) = \dfrac{x}{x^2 + y^2} + i\dfrac{y}{x^2 + y^2}$

In Problems 9–16, (**a**) use Theorem 3.3.2 to show that the given function is analytic in an appropriate domain, and (**b**) use (9) or (11) to find the derivative of the function in the domain.

**9.** $f(z) = e^{-x}\cos y - ie^{-x}\sin y$

**10.** $f(z) = x + \sin x \cosh y + i(y + \cos x \sinh y)$

**11.** $f(z) = e^{x^2-y^2}\cos 2xy + ie^{x^2-y^2}\sin 2xy$

**12.** $f(z) = 4x^2 + 5x - 4y^2 + 9 + i(8xy + 5y - 1)$

**13.** $f(z) = \dfrac{x-1}{(x-1)^2 + y^2} - i\dfrac{y}{(x-1)^2 + y^2}$

**14.** $f(z) = \dfrac{x^3 + xy^2 + x}{x^2 + y^2} + i\dfrac{x^2y + y^3 - y}{x^2 + y^2}$

**15.** $f(z) = \dfrac{\cos\theta}{r} - i\dfrac{\sin\theta}{r}$

**16.** $f(z) = 5r\cos\theta + r^4\cos 4\theta + i(5r\sin\theta + r^4\sin 4\theta)$

In Problems 17–20, find real constants $a$, $b$, $c$, and $d$ so that the given function is entire.

**17.** $f(z) = 3x - y + 5 + i(ax + by - 3)$

**18.** $f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$

**19.** $f(z) = 1 + 5r\cos\theta + ar\sin\theta + i\,(3 + br\sin\theta + r\cos\theta)$

**20.** $f(z) = r^a\cos 2\theta + br\sin\theta + i\left(cr^2\sin\theta\cos\theta + r\cos\theta\right)$

In Problems 21–24, (**a**) show that the given function is not analytic at any point but is differentiable along the indicated curve(s), and (**b**) use (9) or (11) to find the derivative of the function on the curve(s).

**21.** $f(z) = x^2 + y^2 + 2ixy$; $x$-axis

**22.** $f(z) = 3x^2y^2 - 6ix^2y^2$; coordinate axes

**23.** $f(z) = x^3 + 3xy^2 - x + i(y^3 + 3x^2y - y)$; coordinate axes

**24.** $f(z) = x^2 - x + y + i(y^2 - 5y - x)$; $y = x + 2$

**25.** In Section 2.1 we defined the complex exponential function $f(z) = e^z$ in the following manner $e^z = e^x\cos y + ie^x\sin y$.

(**a**) Show that $f(z) = e^z$ is an entire function.

(**b**) Show that $f'(z) = f(z)$

**26.** In Section 3.1 we defined the principal branch of the complex square root in the following manner $z^{1/2} = \sqrt{r}e^{i\theta/2}$, $-\pi < \theta < \pi$.

(**a**) Use (10) to show that $f_1(z) = z^{1/2}$ is analytic in the domain $-\pi < \theta < \pi$.

(**b**) Use (11) to show that $f_1'(z) = \frac{1}{2}z^{-1/2}$ in this domain.

## Focus on Concepts

**27.** Suppose $f(z)$ is analytic. Show that $|f'(z)|^2 = u_x^2 + v_x^2 = u_y^2 + v_y^2$.

**28.** Suppose $u(x, y)$ and $v(x, y)$ are the real and imaginary parts of an analytic function $f$. Can $g(z) = v(x, y) + iu(x, y)$ be an analytic function? Discuss and defend your answer with sound mathematics.

**29.** Suppose $f(z)$ is entire. Use the Cauchy-Riemann equations to show that the function $g(z) = \overline{f(\overline{z})}$ is entire.

**30.** If $f(z)$ and $\overline{f(z)}$ are both analytic in a domain $D$, then what can be said about $f$ throughout $D$?

**31.** In this problem you are guided through the start of the proof of the proposition:

*If $f$ is analytic in a domain $D$, and $|f(z)| = c$, where $c$ is constant, then $f$ is constant throughout $D$.*

**Proof** We begin with the hypothesis that $|f(z)| = c$. If $f(z) = u(x, y) + iv(x, y)$ then $|f(z)|^2 = c^2$ is the same as $u^2 + v^2 = c^2$. The partial derivatives of the last expression with respect to $x$ and $y$ are, respectively,

$$2uu_x + 2vv_x = 0 \quad \text{and} \quad 2uu_y + 2vv_y = 0.$$

Complete the proof by using the Cauchy-Riemann equations to replace $v_x$ and $v_y$ in the last pair of equations. Then solve for $u_x$ and $u_y$ and draw a conclusion. Use the Cauchy-Riemann equations again and solve for $v_x$ and $v_y$.

**32.** If $f(z) = \overline{z}/z$, then show that $|f(z)|$ is constant in the domain defined by $z \neq 0$. Explain why this does not contradict Theorem 3.3.3(*i*).

**33.** Suppose $g(x)$ is a twice differentiable real function and $f$ is a complex function defined by $f(z) = g(x) + iy^2$. Determine where $f(z)$ is differentiable and where it is analytic. Where did you use the hypothesis that $g$ is twice differentiable?

**34.** Find a nonconstant complex function $f$ that is nowhere analytic but is differentiable on the unit circle $|z| = 1$. [*Hint*: Consider using a function $f$ similar to the one in Problem 33.]

**35.** In this problem you are guided through the start of the proof of the proposition:

*If $f$ is analytic in a domain $D$, and $f'(z) = 0$, then $f$ is constant throughout $D$.*

**Proof** We begin with the hypothesis that $f$ is analytic in $D$ and hence it is differentiable throughout $D$. Hence by (9) of this section and the assumption that $f'(z) = 0$ in $D$, we have $f'(z) = \dfrac{\partial u}{\partial x} + i\dfrac{\partial v}{\partial x} = 0$. Now complete the proof.

**36.** Use the Cauchy-Riemann equations to prove that if $f$ is analytic in a domain $D$, and if $f'(z)$ is a constant, then $f(z)$ must be a complex linear function throughout $D$.

**37.** Use the proposition in Problem 35 to show that if $f$ and $g$ are analytic and $f'(z) = g'(z)$, then $f(z) = g(z) + c$, where $c$ is a constant. [*Hint*: Form $h(z) = f(z) - g(z)$.]

**38.** Suppose $x = r\cos\theta$, $y = r\sin\theta$, and $f(z) = u(x, y) + iv(x, y)$. Show that

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta, \qquad \frac{\partial u}{\partial \theta} = -\frac{\partial u}{\partial x}r\sin\theta + \frac{\partial u}{\partial y}r\cos\theta \tag{12}$$

and

$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x}\cos\theta + \frac{\partial v}{\partial y}\sin\theta, \qquad \frac{\partial v}{\partial \theta} = -\frac{\partial v}{\partial x}r\sin\theta + \frac{\partial v}{\partial y}r\cos\theta. \tag{13}$$

Now use (1) in the foregoing expressions for $v_r$ and $v_\theta$. By comparing your results with the expressions for $u_r$ and $u_\theta$, deduce the Cauchy-Riemann equations in polar coordinates given in (10).

**39.** Suppose the function $f(z) = u(r, \theta) + iv(r, \theta)$ is differentiable at a point $z$ whose polar coordinates are $(r, \theta)$. Solve the two equations in (12) for $u_x$ and then solve the two equations in (13) for $v_x$. Then show that the derivative of $f$ at $(r, \theta)$ is

$$f'(z) = (\cos\theta - i\sin\theta)\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right) = e^{-i\theta}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right).$$

**40.** Consider the function

$$f(z) = \begin{cases} 0, & z = 0 \\ \dfrac{z^5}{|z^4|}, & z \neq 0. \end{cases}$$

(**a**) Express $f$ in the form $f(z) = u(x, y) + iv(x, y)$,

(**b**) Show that $f$ is not differentiable at the origin.

(**c**) Show that the Cauchy-Riemann equations are satisfied at the origin. [*Hint*: Use the limit definitions of the partial derivatives $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, and $\partial v/\partial y$ at $(0, 0)$.]

## 3.4 Harmonic Functions

**INTRODUCTION** In Section 5.5 we shall see that when a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic at a point $z$, then *all* the derivatives of $f$: $f'(z)$, $f''(z)$, $f'''(z)$, ... are also analytic at $z$. As a consequence of this remarkable fact, we can conclude that *all* partial derivatives of the real functions $u(x, y)$ and $v(x, y)$ are continuous at $z$. From the continuity of the partial derivatives we then know that the second-order mixed partial derivatives are equal. This last fact, coupled with the Cauchy-Riemann equations, will be used in this section to demonstrate that there is a connection between the real and imaginary parts of an analytic function $f(z) = u(x, y) + iv(x, y)$ and the second-order partial differential equation

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0. \tag{1}$$

This equation, one of the most famous in applied mathematics, is known as **Laplace's equation** in two variables. The sum $\dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial^2\phi}{\partial y^2}$ of the two second-order partial derivatives in (1) is denoted by $\nabla^2\phi$ and is called the **Laplacian** of $\phi$. Laplace's equation (1) is then abbreviated as $\nabla^2\phi = 0$.

### Harmonic Functions

A solution $\phi(x, y)$ of Laplace's equation (1) in a domain $D$ of the plane is given a special name.

**DEFINITION 3.4.1 Harmonic Functions**

A real-valued function $\phi$ of two real variables $x$ and $y$ that has continuous first and second-order partial derivatives in a domain $D$ and satisfies Laplace's equation is said to be **harmonic** in $D$.

Harmonic functions are encountered in the study of temperatures and potentials. These applications are introduced in Section 3.5. In the next theorem we see that both the real and the imaginary parts of an analytic function are harmonic.

**THEOREM 3.4.1 Harmonic Functions**

Suppose the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain $D$. Then the functions $u(x, y)$ and $v(x, y)$ are harmonic in $D$.

**Proof** Assume $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain $D$ and that $u$ and $v$ have continuous second-order partial derivatives in $D$.* Since $f$ is analytic, the Cauchy-Riemann equations are satisfied at every point $z$. Differentiating both sides of $\partial u/\partial x = \partial v/\partial y$ with respect to $x$ and differentiating both sides of $\partial u/\partial y = -\partial v/\partial x$ with respect to $y$ give, respectively,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}. \tag{2}$$

With the assumption of continuity, the mixed partials $\partial^2 v/\partial x \partial y$ and $\partial^2 v/\partial y \partial x$ are equal. Hence, by adding the two equations in (2) we see that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{or} \quad \nabla^2 u = 0.$$

This shows that $u(x, y)$ is harmonic.

Now differentiating both sides of $\partial u/\partial x = \partial v/\partial y$ with respect to $y$ and differentiating both sides of $\partial u/\partial y = -\partial v/\partial x$ with respect to $x$, give, in turn, $\partial^2 u/\partial y \partial x = \partial^2 v/\partial y^2$ and $\partial^2 u/\partial x \partial y = -\partial^2 v/\partial^2 x$. Subtracting the last two equations yields $\nabla^2 v = 0$. ☰

**EXAMPLE 1 Harmonic Functions**

The function $f(z) = z^2 = x^2 - y^2 + 2xyi$ is entire. Therefore, the functions $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$ are necessarily harmonic in any domain $D$ of the complex plane. This is also easy to verify directly:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2 = 0 \quad \text{and} \quad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 + 0 = 0.$$ ☰

**Harmonic Conjugate Functions** We have just shown that if a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain $D$, then its real and imaginary parts $u$ and $v$ are necessarily harmonic in $D$. Now suppose $u(x, y)$ is a given real function that is known to be harmonic in $D$. If it is possible to find another real harmonic function $v(x, y)$ so that $u$ and $v$ satisfy the Cauchy-Riemann equations throughout the domain $D$, then the function $v(x, y)$ is called a **harmonic conjugate** of $u(x, y)$. By combining the functions as $u(x, y) + iv(x, y)$ we obtain a complex function that is analytic in $D$.

**EXAMPLE 2 Harmonic Conjugate**

(**a**) Verify that the function $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic in the entire complex plane.

(**b**) Find a harmonic conjugate of $u$.

*The continuity of the second-order partial derivatives of $u$ and $v$ is not part of the hypothesis of the theorem. This fact will be proved in Chapter 5.

**Solution** (**a**) From the partial derivatives

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial^2 u}{\partial x^2} = 6x, \quad \frac{\partial u}{\partial y} = -6xy - 5, \quad \frac{\partial^2 u}{\partial y^2} = -6x$$

we see that $u$ satisfies Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x - 6x = 0.$$

(**b**) Since a harmonic conjugate $v$ must satisfy the Cauchy-Riemann equations $\partial v/\partial y = \partial u/\partial x$ and $\partial v/\partial x = -\partial u/\partial y$, we must have

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2 \quad \text{and} \quad \frac{\partial v}{\partial x} = 6xy + 5. \tag{3}$$

Partial integration of the first equation in (3) with respect to the variable $y$ gives $v(x, y) = 3x^2y - y^3 + h(x)$. The partial derivative with respect to $x$ of this last equation is

$$\frac{\partial v}{\partial x} = 6xy + h'(x).$$

When this result is substituted into the second equation in (3) we obtain $h'(x) = 5$, and so $h(x) = 5x + C$, where $C$ is a real constant. Therefore, the harmonic conjugate of $u$ is $v(x, y) = 3x^2y - y^3 + 5x + C$. ☰

In Example 2, by combining $u$ and its harmonic conjugate $v$ as $u(x, y) + iv(x, y)$, the resulting complex function

$$f(z) = x^3 - 3xy^2 - 5y + i(3x^2y - y^3 + 5x + C)$$

is an analytic function throughout the domain $D$ consisting, in this case, of the entire complex plane. In Example 1, since $f(z) = z^2 = x^2 - y^2 + 2xyi$ is entire, the real function $v(x, y) = 2xy$ is a harmonic conjugate of $u(x, y) = x^2 - y^2$. See Problem 22 in Exercises 3.4.

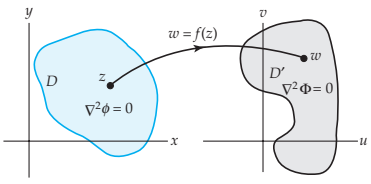



**FIGURE 3.4.1** A solution of Laplace's equation in $D$ is found by solving it in $D'$.

### *Remarks* COMPARISON WITH REAL ANALYSIS

In this section we have seen if $f(z) = u(x, y) + iv(x, y)$ is an analytic function in a domain $D$, then both functions $u$ and $v$ satisfy $\nabla^2\phi = 0$ in $D$. There is another important connection between analytic functions and Laplace's equation. In applied mathematics it is often the case that we wish to solve Laplace's equation $\nabla^2\phi = 0$ in a domain $D$ in the $xy$-plane, and for reasons that depend in a very fundamental manner on the shape of $D$, it simply may not be possible to determine $\phi$. *But* it may be possible to devise a special analytic mapping $f(z) = u(x, y) + iv(x, y)$ or

$$u = u(x, y), \quad v = v(x, y), \tag{4}$$

from the $xy$-plane to the $uv$-plane so that $D'$, the image of $D$ under (4), not only has a more convenient shape but the function $\phi(x, y)$ that satisfies Laplace's equation in $D$ also satisfies Laplace's equation in $D'$. We then solve Laplace's equation in $D'$ (the solution $\Phi$ will be a function of $u$ and $v$) and then return to the $xy$-plane and $\phi(x, y)$ by means of (4). This *invariance* of Laplace's equation under the mapping will be utilized in Chapters 4 and 7. See FIGURE 3.4.1.

## EXERCISES 3.4

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1–10, (**a**) verify that the given function $u$ is harmonic in an appropriate domain $D$, (**b**) find $v(x, y)$, a harmonic conjugate of $u$, and (**c**) form the corresponding analytic function $f(z) = u + iv$.

**1.** $u(x, y) = x$

**2.** $u(x, y) = 2x - 2xy$

**3.** $u(x, y) = x^2 - y^2$

**4.** $u(x, y) = x^3 - 3xy^2$

**5.** $u(x, y) = \log_e(x^2 + y^2)$

**6.** $u(x, y) = \cos x \cosh y$

**7.** $u(x, y) = e^x(x \cos y - y \sin y)$

**8.** $u(x, y) = -e^{-x} \sin y$

**9.** $u(x, y) = \dfrac{x}{x^2 + y^2}$

**10.** $u(x, y) = \arctan(-y/x)$

In Problems 11 and 12, verify that the given function $u$ is harmonic in an appropriate domain $D$. Find a harmonic conjugate $v$ and find an analytic function $f(z) = u + iv$ satisfying the indicated condition.

**11.** $u(x, y) = xy + x + 2y - 5$; $f(2i) = -1 + 5i$

**12.** $u(x, y) = 4xy^3 - 4x^3y + x$; $f(1 + i) = 1 + 4i$

**13.** (**a**) Show that $v(x, y) = \dfrac{x}{x^2 + y^2}$ is harmonic in a domain $D$ not containing the origin.

(**b**) Find a function $f(z) = u(x, y) + iv(x, y)$ that is analytic in domain $D$.

(**c**) Express the function $f$ found in part (b) in terms of the symbol $z$.

**14.** Suppose $f(z) = u(r, \theta) + iv(r, \theta)$ is analytic in a domain $D$ not containing the origin. Use the Cauchy-Riemann equations (10) of Section 3.3 in the form $ru_r = v_\theta$ and $rv_r = -u_\theta$ to show that $u(r, \theta)$ satisfies Laplace's equation in polar coordinates:

$$r^2 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0. \tag{5}$$

In Problems 15 and 16, (**a**) use (5) to verify that the given function $u$ is harmonic in a domain $D$ not containing the origin, (**b**) use equations (10) of Section 3.3 to find $v$, a harmonic conjugate of $u$, and (**c**) form the corresponding analytic function $f(z) = u + iv$.

**15.** $u(r, \theta) = r^3 \cos 3\theta$

**16.** $u(r, \theta) = \dfrac{10r^2 - \sin 2\theta}{r^2}$

### Focus on Concepts

**17.** (**a**) Verify that $u(x, y) = e^{x^2 - y^2} \cos 2xy$ is harmonic in an appropriate domain $D$.

(**b**) Find a harmonic conjugate $v$ and find an analytic function $f(z) = u + iv$ satisfying $f(0) = 1$. [*Hint*: When integrating, think of reversing the product rule.]

**18.** Suppose that $f(z)$ is an entire function, $\text{Re}(f(z)) = y^3 - 3x^2y + 2$, and $f(i) = 2 - i$. Find $f(1 + i)$.

**19.** Express the function $f$ found in Problem 11 in terms of the symbol $z$.

**20.** Express the function $f$ found in Problem 16 in terms of the symbol $z$.

**21.** (**a**) Show that $\phi(x, y, z) = \dfrac{1}{\sqrt{x^2 + y^2 + z^2}}$ is harmonic, that is, satisfies Laplace's equation $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$ in a domain $D$ of space not containing the origin.

(**b**) Is the two-dimensional analogue of the function in part (a), $\phi(x, y) = \dfrac{1}{\sqrt{x^2 + y^2}}$, harmonic in a domain $D$ of the plane not containing the origin?